Skill Diversification Beyond High-Paying Jobs*

Siqi Han

Chinese University of Hong Kong

Siwei Cheng

New York University

*Authors contributed equally to this work. This work received funding from the Russell Sage Foundation, the Washington Center for Equitable Growth, and New Faculty Fund from the Chinese University of Hong Kong. Earlier versions of this paper were presented at the American Sociological Association's 2023 Annual Meeting, the Future of Work and Occupations workshop at NYU Department of Sociology, the 2024 Sociological Science Conference at Duke University,the 2025 RC 28 Conference at UCLA, the Sociology Colloquium at Stony Brook University, Sociology Seminar at Nuffield College at the University of Oxford, the Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) in Nuremberg, the Department of Sociology at Princeton University, the Department of Sociology at the University of Wisconsin-Madison, and the CAREER Workshop at the University of Amsterdam. Direct correspondence to Siqi Han, Assistant Professor of Sociology at the Chinese University of Hong Kong (siqihan@cuhk.edu.hk) and Siwei Cheng, Associate Professor of Sociology at New York University (siwei.cheng@nyu.edu).

Skill Diversification Beyond High-Paying Jobs

Abstract

An emerging literature examines the evolving skill profiles in the contemporary labor market, but remains limited in three key respects. It emphasizes occupational over job-level dynamics, focuses on high-paying jobs while overlooking low-paying ones, and ignores interrelationships among skills across the labor market. We advance this literature with a Skill Embedding framework that maps skills onto a multidimensional space to capture their interconnections. Using large-scale job postings data, we construct within-job skill diversity measures and link them to nationally-representative surveys. We find that the increase in within-job skill diversity in service jobs is three times as large as that in management jobs. In both professional and personal service occupations, diversification is largely driven by lower-status workers, and returns are lower in personal service jobs than in professional occupations. These findings reveal overlooked skill dynamics in low-wage work and demonstrate the value of a system-wide, distributional approach to analyzing skill relationships.

Introduction

The contemporary labor market is characterized by a dynamic and complex demand for skills that continuously evolve in response to technological advancements (Autor et al., 2008; Goldin and Katz, 2010; Liu and Grusky, 2013) and changes in institutional and organizational contexts (Altonji, 2010; Baron and Bielby, 1980; Tomaskovic-Devey and Avent-Holt, 2019). Over the last two decades, research in work and occupation has moved from a simplistic binary view of skills, categorizing workers as either "high-" or "low-" skilled, toward a more comprehensive understanding that recognizes the co-evolving nature of skill requirements across *multiple* dimensions (e.g. Autor et al., 2008; Deming, 2017; Goldin and Katz, 2010; Liu and Grusky, 2013). In response to the growing variety of skills in the labor market, an emerging line of work has turned its focus towards the *combination* of different skill dimensions. For example, Börner et al. (2018) depicted how hard and soft skills become more coupled for workers in technology; Hosseinioun et al. (2025) discovered a nested structure of skills in job requirements, which follow a hierarchy such that specialized expertise is embedded within broader layers of general skills; and Anderson (2017) analyzed an online freelance labor market, showing that workers with diverse skills earn higher wages and that those who use their diverse skills in combination earn the highest wages of all.

All this recent evidence highlighted the importance of *skill diversity*, shifting scholarly attention from the level of skills to the diversity of skills demanded by employers in the job market. However, much of this research to date has focused on the highly skilled segment of the labor market, examining professional occupations such as tech workers and scientists, or those in management roles. In contrast, discussions about skill diversity seldom extend to low-wage workers in personal service and care occupations. These jobs are often portrayed as occupying the low-skill periphery of the labor market and assumed to require relatively simple skill sets. To put it simply, when the occupants of a job are lowly-paid, the work that they do is also often perceived as low-skilled (Warhurst et al., 2017; Bryson, 2017; Vallas, 1990; Attewell, 1990).

But must these low-wage jobs necessarily lack skill diversity? We challenge this common perception. We comprehensively investigate skill diversification across the entire occupational spectrum. If technological change and employment expansion have driven skill diversification in management and professional jobs, could a similar trend be unfolding in rapidly growing low-wage occupations? To explore this, we focus on personal care and service jobs, which have seen employment growth on par with high-paying occupations in recent decades (Autor and Dorn, 2013). Beyond technological and employment changes, skill profiles in personal service jobs are shaped by two additional forces. First, the commodification of care work and increasingly diverse consumer preferences may be expanding skill demands in these jobs (Duffy, 2011). Second, workers in these sectors have precarious yet flexible working conditions, leading them to be able to absorb diverse job tasks that professional workers from other occupations deploy to them (Reckrey et al., 2019), therefore diversifying the skills of the jobs they hold. Looking beyond high-paying jobs to examine evolving skill profiles provides a fuller picture of how technological, institutional, and organizational forces have reshaped the workplace.

Another important limitation of prior research is its predominant focus on occupation-level skill diversity, overlooking variation at the *job level*. Without looking within occupational boundaries, it remains unclear whether skill diversification occurs between different jobs in the same occupation or within individual jobs. Our study builds on an emerging line of research in the sociology of work that emphasizes *job-level* dynamics (Avent-Holt et al., 2020; Tomaskovic-Devey and Avent-Holt, 2019). Jobs represent the practical context where specific skill sets are combined and utilized in the workplace (Hauser and Warren, 1997). They also serve as the platform for negotiations between workers and employers over responsibilities and compensation (Alabdulkareem et al., 2018; Baron and Bielby, 1986; Martin-Caughey, 2021). Lacking sufficient job-level data has significantly limited prior research's ability to examine job-level skill profiles within detailed occupations. Our study leverages large-scale job postings data, which presents a unique opportunity to advance the literature with a focus on job-level analysis.

Furthermore, a close examination of patterns and trends in skill diversity in the contemporary workplace calls for a relational, system-wide, and context-dependent perspective. In this study, we build on insights from recent work on skills and knowledge to conceptualize skill diversity in terms of the *atypicality of skill combinations embedded in a broader skill system* (Alabdulkareem et al., 2018; Uzzi et al., 2013). From this perspective, jobs that combine skills rarely observed together exhibit higher levels of skill diversity. Skill diversity, in this view, is not intrinsic to a job alone but depends on the distribution of skills across the labor market. Accordingly, understanding skill diversity requires foregrounding the full constellation of skills—and their interconnections—as its theoretical foundation. Because it is inherently embedded in the skill space itself, this approach also flexibly adapts to the shifting topology of skills over time and across contexts.

In the meantime, analyzing job-level skill diversity poses significant methodological challenges. When analyzing multi-dimensional skills, prior studies tend to treat individual skills as discrete dimensions and rely on measures that simply count the number of skills. However, merely counting the number of skills required for a job may not accurately reflect the diversity of skills involved: jobs that demand a small number of highly dissimilar skills (e.g., "programming" and "communication and motivation") may exhibit a more diverse skill set than jobs requiring a larger number of similar skills (e.g., “SQL database management,” “python,” and “linear algebra”). Therefore, to effectively measure skill diversity requires assessing the relative "distances" among multiple skills. To address this challenge, we introduce an analytical framework termed *Skill Embedding*, which maps a broad array of skills onto a multidimensional vector space based on the frequency of their co-occurrence within the same job and with other skills. This framework allows us to incorporate the varying distances between skill dimensions into our measurement of skill diversity.

Employing this skill embedding framework, we extract detailed skill requirements from large-scale job postings data to construct measures of job-level skill diversity. These measures are then matched to nationally representative surveys to examine their trends and earnings returns. Our analysis reveals three main findings. First, within-job skill diversity has increased in management and professional occupations, but the rise in skill diversity is even larger in the personal care and service sector at the lower end of the earnings spectrum. Second, within professional and personal service occupations, the trend toward greater skill diversification is driven primarily by lower-status rather than higher-status workers. Third, after accounting for a range of demographic and educational factors, workers in personal service jobs receive lower earnings returns to within-job skill diversity compared to those in management and professional occupations. We discuss the implications of our findings at the end of this article.

Conceptualizing Skill Diversity from a Relational Perspective

Skill diversity in the workplace can be conceptualized in different ways. The functional perspective defines skill diversity in the workplace by the number of functionally distinct skills a job requires. It considers a skill set to be more diverse when it spans multiple domains of activity. For instance, a surgeon's skill set can be viewed as diverse, because it involves both performing surgical procedures and communicating with patients in the case of surgeons. Yet beyond functional distinctiveness, skill diversity can also be understood as a relational feature of a skill set: skill sets differ not only in the functional scope of their constituent skills, but also in how (a)typical particular skill combinations are across the labor market (Alabdulkareem et al., 2018; Börner et al., 2018). From this perspective, a skill set is considered diverse insofar as it *combines skills that are rarely observed together in the same or similar jobs*, as such unusual combinations contribute to the diversity of skill combinations in the labor market. A job that involves an uncommon pairing of skills signals greater skill diversity in a given labor market context, because it bundles together capabilities that are rarely combined in a single job. For example, the skill set of a digital humanities researcher is diverse because it combines humanistic interpretive expertise with computational skills—a pairing that is both functionally diverse and remains rare in the contemporary labor market; in contrast, the skill set of a user experience engineer, which may combine focus group surveying and programming, is functionally diverse but not as rare.[1] An analogy to ecological diversity is helpful here. In the study of ecological communities, one can focus on functional diversity—that is, the extent to which species perform different roles within an ecosystem. But one can also emphasize relational diversity: a community may be considered diverse when it brings together species that rarely co-occur elsewhere, because such unusual combinations contribute to the diversity of species assemblages across the broader ecosystem. Both dimensions of diversity contribute to our understanding of the system, though they do so from distinct and complementary perspectives.

While we are not advocating for replacing one conceptualization of skill diversity with another—and indeed these measures are not mutually exclusive in the empirical world—we argue that the relational perspective provides distinct and important insights into labor market skill profiles that are not captured by the functional perspective alone. In particular, the relational perspective has two key implications. First, at a theoretical level, it requires adopting a *system-wide, distributional view* on the labor market. Whereas measuring the functional diversity of a skill set requires assessing the functional similarity among the skills contained within a given job, evaluating the atypicality of a skill set cannot be accomplished by examining that job in isolation. Instead, atypicality is inherently relational: it depends on how the same skills co-occur across *other* jobs throughout the labor market. Returning to our earlier example of a digital humanities researcher in the previous paragraph, this position can be recognized as combining an unusual set of skills only when we observe that humanistic and computational skills are rarely bundled together across the broader distribution of jobs.

[1] While related, our definition of skill diversity is distinct from the concept of skill complementarity. Skill complementarity typically refers to how well different skills enhance each other's productivity or performance when used together. In contrast, our definition of skill diversity emphasizes the breadth and atypicality of skill combinations within a job, regardless of whether those skills are functionally complementary. A job may exhibit high skill diversity by combining rarely co-occurring or cross-domain skills (e.g., creative writing and data analytics), even if there are limited productivity gains from such combinations.

Second, at a substantive level, the relational perspective allows measures of skill diversity to be *context-dependent and time-varying*, as it can flexibly adapt as the structure of skill pairings in the labor market evolves over time. Whereas empirical analyses adopting the functional perspective often rely on pre-specified and largely static metrics of skill similarity, the relational approach uncovers skill-to-skill diversity inductively through observed co-occurrence patterns.[2] This approach is particularly well suited to capturing rapid temporal changes in the labor market context—a defining feature of the contemporary economy. For example, journalistic reporting and data-analytic skills historically occupied largely separate regions of the labor market and would therefore be considered as distant skills; today, however, many journalistic positions require the integration of reporting with data competencies, leading to an increase in the similarity between these skills. This emerging trend of a new type of journalism work will not reveal itself unless our measure explicitly accounts for its context-dependent and time-varying nature. As we demonstrate in this study, such temporal shifts can be directly documented by applying a relational perspective to large-scale, time-stamped data on job-level skill requirements.

Skill Diversity: Two Critical Gaps in the Literature

Although skill diversity has received substantial attention in labor market research, two critical gaps remain. First, most studies rely on occupation-level, rather than job-level analysis. Second, they focus primarily on high-paying occupations—such as managerial and professional jobs–and offer little insight into the skill profiles of low-wage work. In the next two subsections, we elaborate on these two arguments.

Beyond Occupations: Differentiating Within- and Between-job Skill Diversity

In stratification research, occupations have traditionally been considered as groupings that bundle skills into skill profiles (Blau et al., 1978; Hout, 1984; Liu and Grusky, 2013; Wee-

[2]We describe the methodology in detail in later sections.

den and Grusky, 2005). Comprehensive, large-scale data sets, such as the Occupational Information Network (O*NET), and more recently, the verbatim text responses provided by respondents (Martin-Caughey, 2021), have empowered research to assess the heterogeneity of skill profiles for different occupations. Research has looked into the evolving requirements for *specific* skill dimensions for different occupations (e.g. Autor et al., 2008; Cheng et al., 2019; Deming, 2017; Goldin and Katz, 2010; Liu and Grusky, 2013), and more recently, the diversity of skill profiles at the occupation level (Eggenberger et al., 2018; Hénaut et al., 2023; Martin-Caughey, 2021).

Yet, at the same time, research in the sociology of work increasingly calls for a shift in focus from occupational categories to individual jobs. While occupations are usually used to observe how particular skills are bundled together, jobs are where these skills are practically applied by individual workers in expectation of economic compensation (Hauser and Warren, 1997). Jobs represent "the actualization of particular tasks requiring particular skills within a given workplace." Avent-Holt et al., 2020, p. 3. Many established processes of labor market stratification, such as between-job segregation, opportunity hoarding, skill-upgrading, significantly influence people's jobs rather than operate at the level of their occupations (Baron and Bielby, 1986; Baron and Newman, 1990; Kim and Sakamoto, 2008; Martin-Caughey, 2021; Tomaskovic-Devey and Avent-Holt, 2019). Jobs are also critical venues for the negotiation between workers and employers regarding responsibilities, duties, and compensation of individual roles. Job seekers focus on specific positions rather than broader occupational categories when applying for jobs. Workers develop skills and task-specific human capital through the daily execution of their job roles (Gibbons and Waldman, 2004; Kalleberg and Mouw, 2018).

In this study, we build on the growing focus on jobs to underscore the need for a systematic analysis of *job-level skill profiles*. To illustrate the conceptual distinction between occupation- and job-level skill profiles, Figure 1 presents a visualization of the multi-layered nature of skill diversity. Here, in keeping with the analytic framework to be introduced later, we use vectors to represent skills. Vectors pointing to closer directions represent more similar skills. In this figure, the top layer represents the overall labor market, encompassing the entire spectrum of workplace skills. The middle layer segments the market into occupations and shows the skill profiles associated with each occupation. The arrows indicate skills required for all jobs in an occupation, without breaking them down into specific jobs. The bottom layer highlights the skill profiles specific to individual job positions: within each occupation, the four skill vectors are further subdivided into two jobs. Within each job, Jobs 1 and 2 combine a more similar array of skills than the relatively heterogeneous skill sets found within Jobs 3 and 4.

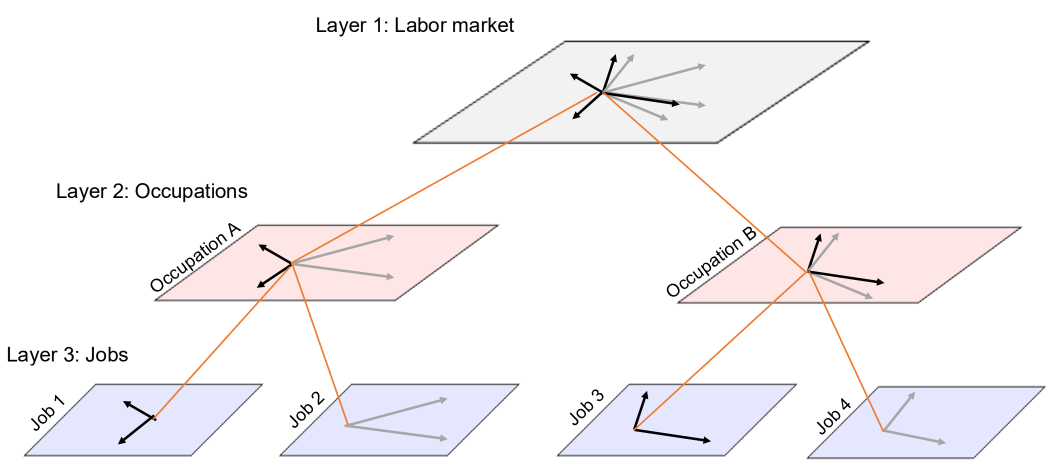


Figure 1: Distinguishing Multiple Layers of Skill Diversity

Note: The arrows represent skill vectors. Skill vectors that are closer to each other represent more similar skills.

Alt Text: Three stacked planes illustrate the multi-layered structure of skill diversity. The top plane represents the labor market with skill vectors. The middle plane splits into two occupations, each with four skill vectors. The bottom plane subdivides each occupation into two jobs, showing skill vectors at the job level.

Distinguishing between occupation- and job-level skill profiles is not merely a matter of analytical granularity—in fact, it reveals substantively distinct forms of skill diversification that remain hidden when skill profiles are aggregated at the occupational level. Figure 2 illustrates this point. Here, each chess piece represents a worker in a specific job, and the surrounding logos denote the distinct skills required for that job. When two workers share the same chess icon, they possess the same set of skills. Now, consider a scenario where we only have occupation-level skill data: Occupations A and B will appear identical, as both require four distinct skills on the whole. However, when we zoom in and examine job-level data, a crucial distinction emerges: in Occupation A, each worker possesses all four skills, indicating high within-job skill diversity but no variation across jobs. In contrast, each worker in Occupation B specializes in a single, unique skill, indicating low within-job skill diversity but high between-job diversity. If we only consider skill diversity at the occupational level, we risk conflating these two different scenarios.

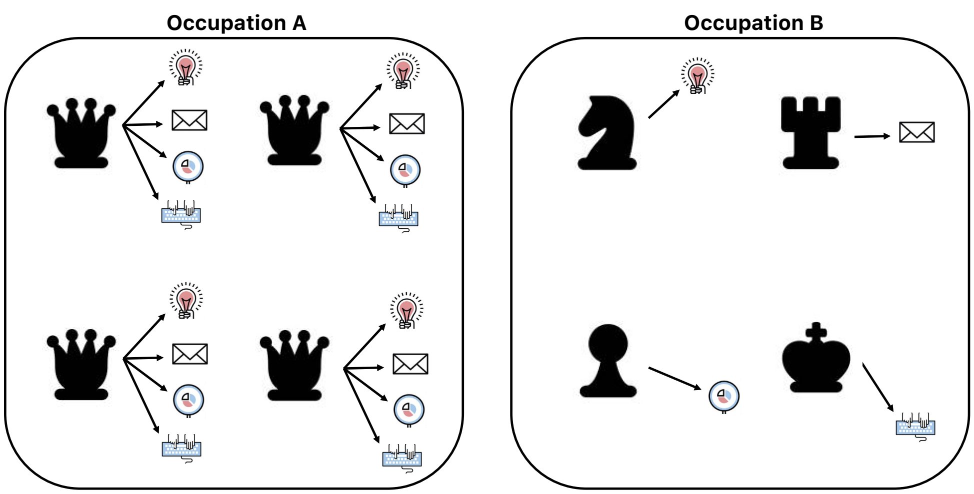


Figure 2: Contrasting Within-job versus Between-job Skill Diversity

Note: Each chess piece represents a worker in a specific job role; the logos surrounding each piece denote the distinct skills required for that job. When two workers share the same chess icon, it indicates that they possess the same set of skills.

Alt Text: Two boxed panels compare Occupation A and Occupation B. Occupation A contains four identical queen chess pieces, each surrounded by four different skill icons. Occupation B contains four different chess pieces, each surrounded by a single unique skill icon. Both occupations require four distinct skills in total.

Beyond being a critical analytical concept, skill profiles at the job level have profound substantive and theoretical importance. To begin, the skill profile of a job position communicates expectations about a job's scope to potential candidates. Expanding a job's skill range broadens its scope, whereas defining a unique skill set distinguishes it from others. Such requirements motivate potential job seekers to tailor their skill portfolios to meet employer demands (Börner et al., 2018), and the linkage between skills trained in the educational system and those required on the job further influences earnings (Bol et al., 2019; DiPrete and Chae, 2023; Han et al., 2023). The constant evolution of job-level skill profiles, which may include the integration of new specialties or the bridging of previously isolated domains (Burt, 1995), is instrumental in reinforcing social closure, creating a hierarchy of status and power among different jobs (Baron and Newman, 1990; Parkin, 1982; Tomaskovic-Devey and Avent-Holt, 2019; Weeden, 2002).

More specifically, as our example in Figure 2 shows, analyzing skill diversity at the job level allows us to explore the distinction between generalists versus specialists and their evolving roles in the labor market. Specialists delve deeply into a particular area of their profession (e.g., jobs in Occupation B in Figure 2), while generalists spread their expertise across a wide array of domains (e.g., jobs in Occupation A in Figure 2). While the conventional wisdom might predict positive economic returns to specialists' heightened proficiency in a specific domain, recent research suggests that the current economy seems to be favoring workers who combine skills across domains (Becker, 1962; Brynjolfsson, 2014; Deming, 2017; Jackson, 2025). By extending their knowledge to encompass diverse domains, generalists can bridge gaps between various disciplines such as natural science, social sciences, humanities, and arts, thereby fostering interdisciplinary synergy (Han et al., 2023). This blending of skills is increasingly valued for its potential to drive innovation, as integrating different knowledge areas can spark new ideas and advancements. For example, Deming, 2017 shows that in workplaces increasingly organized around team-based settings, professional workers with stronger social skills face lower collaboration costs, are more effective at leveraging other team members' strengths, become more productive themselves, and adapt more flexibly to changing circumstances. The incorporation of social skills into highly specialized professional jobs, such as software engineers, has contributed to the diversification of their skill sets. These workers are starting to possess not only technical expertise but also interpersonal skills, and such newer combinations benefit their own productivity and earnings.[3]

Further, the range of skills demanded on the job significantly influences the specific forms of human capital that employees develop, which in turn shapes their future labor market prospects (Becker, 1962; Gibbons and Waldman, 2004; Kalleberg and Mouw, 2018). The degree of similarity in skill requirements between two positions is crucial for determining wage outcomes when employees transition between roles. As a result, having a broad skill set helps generalists mitigate labor market risks. In situations of involuntary job displacement, workers with a versatile skill set are better positioned than those with specialized skills (Byun and Raffiee, 2023).[4] Wage losses associated with job displacement are more strongly correlated with changes in skill sets than with transitions across industry or occupational categories (Poletaev and Robinson, 2008). This underscores the significance of maintaining a broader range of skills that can be flexibly transferred across jobs when experiencing career instability.

Despite the significance of job-level skill diversity for labor market research, empirical analyses on this topic remain limited in two ways. First, to date, recent studies have examined only a few specific sectors in the labor market, such as music (Faulkner, 1983), film (Zuckerman et al., 2003), MBAs (Merluzzi and Phillips, 2016), labor union workers (Wilmers, 2020), and freelancers (Anderson, 2017; Leung, 2014). Yet, to our knowledge, no prior research has examined the trends and returns of job-level skill diversity spanning the *entire spectrum of jobs*. This gap is partly due to the lack of detailed job-level skill requirement information in conventional survey data. Recently, the availability of large-scale, online

[3]To use one of Deming's own empirical findings as an example (Deming, 2017, pp. 1599), "I find that employment and wage growth has been particularly strong in occupations with *high math* and *social skill* requirements. In contrast, employment has declined in occupations with high math but low social skill requirements, including many of the STEM jobs shown in Figure I."

[4]Byun and Raffiee, 2023 focuses on skill diversity accumulated across a career, rather than within a single job. However, the advantages of being a generalist should apply regardless of whether diverse skills are acquired across multiple positions or within a single job.

job postings data sets presents a unique opportunity to address this gap. Second, although prior work has made significant progress documenting the *multidimensionality* of skills, they often consider these skill dimensions as discrete domains.[5] However, to capture the diversity of skills, we need to first assess the relative "distance" between them. To address this, we propose a vector representation of skills that captures the relationship between skills. We will describe the details in the methods section.

Beyond High-paying Jobs: Has Skill Diversity Increased in Low-paying Jobs?

Most prior discussions of skill diversification have focused on high-paying jobs. Has skill diversification also emerged across the entire occupational spectrum? We argue that the skill profiles in a rapidly expanding occupational group at the opposite end of the earnings spectrum—namely, personal care and service jobs—may be shaped by forces of technological change, employment expansion, and customer demands diversification, but in ways that are distinct from those seen in management and professional occupations. Below, we outline these forces and specifically illustrate how they influence skill diversification of the lower-paying jobs.

The first driving force is technological transformation (Liu and Grusky, 2013), which brings new skills to both higher- and lower-paying jobs (Dwyer, 2013; Duffy, 2011; Sassen, 2001). The growing emphasis in scientific knowledge on multiple, rather than singular, mechanisms—along with an expanding array of tools for intervening in these mechanisms—has also contributed to the broadening of job tasks over time (Jackson, 2025). As scientific advances continue, many jobs now require a diverse mix of knowledge and skills rather than a single form of expertise, often bringing together skill domains once considered distant within the same job. For lower-paying jobs specifically, on the one hand, AI-assisted services and cyberization are creating a new complementarity (Snower and Goerlich, 2013) between human

[5]Even in the few studies that map skills onto an inter-related network landscape (e.g. Alabdulkareem et al., 2018), the emphasis has largely been placed on the frequency and the degree of co-occurrences between skill pairs rather than the dissimilarity or diversity of skills.

services and the machine, requiring jobs to involve skills related to using these technologies (Huang and Rust, 2018) and lowering the costs of performing single tasks and leaving more room for multitasking. In the personal care and service sector, for example, childcare providers are increasingly adopting AI-based educational tools and apps that support early childhood development and learning; personal trainers are now using big-data-driven fitness apps and wearables that track client progress to provide real-time feedback on client performance. On the other hand, with the adoption of productivity management in the major platform economy, service workers need to be proficient in skills dealing with the platform algorithm. This includes knowing how to log attendance, keeping track of their hours and deadlines, and answering demand calls from customers and supervisors (Lehdonvirta, 2018; Vallas and Schor, 2020). These complex new realities that service workers find themselves in compel them to be more skill-diverse.

The second potential factor is the labor market sector expansion in the context of job polarization — that is, the expansion of upper- and lower-end of the labor market at the expense of shrinking middle-class jobs (Autor et al., 2008; Althobaiti et al., 2022). The recent expansion of the low-wage service sector has been equally dramatic—if not more so—than that of management and professional jobs at the higher end. As shown in Figure S2 and S3 in the Online Supplement, three occupational groups experienced significant expansion between 2010 and 2017, highlighted by the thick blue lines: professional and management jobs at the higher end of the earnings spectrum, and personal care and service jobs at the lower end.[6] As predicted by the niche width theory, an important consequence of the sector expansion of the top and bottom sectors is that companies in these sectors will expand their skill requirements to adapt to rapidly changing environments, thereby increasing their profitability (Carroll, 1995; Hannan and Freeman, 1977). As companies create jobs that combine various skills, promote collaboration, and fill more niches in the labor market,

[6]The numbers shown on this figure represent the percentage change in employment shares across various occupational groups based on data obtained from the American Community Survey. Online Supplement A provides a full list of detailed occupations under the Personal Care and Service category in the SOC 2010 occupation classification.

skill diversity increases. Not only was there continuous polarization between high-paying and low-paying service occupations and across the occupational structure, but also within the occupations (Cheng et al., 2025). The rapid expansion of the low-wage service sector could introduce new service tasks and novel combinations of job skills to these occupations, similar to the impact seen in their high-paying counterparts. The increased influx of workers into these occupations may foster the emergence of new hybrid skill combinations—such as integrating health-support tasks with personal care—thereby increasing skill diversity.

Third, the diversifying customer demands may also drive up the demand for a more diverse skill set for both high- and low-paying jobs. It is not surprising that new managerial tasks or financial products emerge every year to cater to customers' demand for higher-end services. On the other hand, the customer demands in the personal service and care sector are no less complex. Personal care work—including jobs such as childcare, nursing, and therapy—typically involves direct service delivery to clients. These jobs now require a wide range of skills spanning instrumental tasks and affective labor, including both nurturant (e.g., teaching, healthcare) and reproductive work (e.g., cleaning, cooking) (Duffy, 2011; England, 2005). Providing care thus demands diverse person-oriented skills — manual, technical, medical, and emotional (Duffy, 2011; Malhotra, 2022), which endow such work with discretion, task variety, and employees' control (Gatta et al., 2009; Bolton, 2004). For example, hairdressers and beauty specialists must combine grooming, artistic styling, massage, skincare, and personalized communication (Banerjee, 2023).

Skill diversity in personal care jobs is likely to grow as consumers' needs become more complex, requiring multi-skilled workers and organizations capable of addressing challenges across domains (Autor and Dorn, 2013; Dwyer, 2013; Snower and Goerlich, 2013). Rising income inequality at the top (Piketty and Saez, 2006) and the increasing participation of women in professional roles (Blau et al., 2013; England et al., 2020) are expected to further boost demand for personal care services and recognition of caregivers' diverse skills. For example, as middle- and upper-class families invest more in their children (Kornrich and

Furstenberg, 2013; Schneider et al., 2018) and more highly educated mothers remain in the workforce (Bauer and Wang, 2023), childcare workers must now go beyond basic care to provide educational and emotional support, foster learning through play, help children develop coping skills, and communicate effectively with parents (Jackson, 2025). Demographic diversity is another driver for changing customer and rising job skill diversity. With the growing number of immigrants in some neighborhoods and gateway cities, bartenders and servers who possess multiple languages and knowledge of foreign cultures gain unique advantages in providing face-to-face service in a culturally appropriate way (Wilson, 2018). Meanwhile, the increasing commodification of personal care, driven by a rising immigrant workforce and shifting norms that favor professional over familial caregiving (Duffy, 2011; Gonalons-Pons and Marinescu, 2022).

Beyond the mechanisms discussed above, job precarity represents an additional mechanism contributing to skill diversification in lower-paying jobs—one that is largely absent in high-paying jobs. Low-wage workers in personal care and service occupations are increasingly exposed to precarious, low-paying employment characterized by unstable contracts and limited job security (Kalleberg, 2009; Kalleberg, 2011; Schneider and Harknett, 2021). This instability and precarity also enable them to absorb the unmet customer demands that the licensed professionals can no longer adequately handle. For example, with the national nursing staff shortages (Juraschek et al., 2019) and the increasing complexity of patient care needs (Juraschek et al., 2019), more states allow for certain tasks to be delegated from registered nurses to unlicensed personal care providers. These tasks may range from working with people with physical disabilities, food nutrition and meal preparation, to safety and injury prevention, or even medicine administration (Reckrey et al., 2019; Saari et al., 2018). Unlicensed personal care providers may receive training and supervision from registered nurses to perform these tasks, which can increase their fluency in carrying out newly assigned responsibilities. This process, in turn, can increase the job-level skill diversity of personal care. At the same time, firms have increasingly adopted polarized employment strategies, separating workers into core (skilled, stable) and peripheral (less-skilled, temporary) jobs (Kalleberg, 2011). Many personal service jobs fall into the peripheral category. Employers pursuing low-road strategies face weak incentives to retain or invest in workers and instead shift employment risk onto employees. Under these conditions, firms are more likely to expand task expectations and demand broader skill sets from low-wage workers in order to maximize the utilization of their human capital (Barney and Wright, 1998). These employer-driven pressures interact with workers' own motivations to diversify their skills in pursuit of greater autonomy, task variety, and intrinsic job satisfaction (Clegg and Spencer, 2007; Hénaut et al., 2023; Morgeson and Humphrey, 2006; Oldham and Hackman, 2010). Together, these dynamics contribute to increasing within-job skill diversification in low-wage personal care and service occupations.

Skill Embedding: A Framework for Analyzing Skill Diversity

Methodological Challenge: Accounting for the Relationship between Skills

The analysis of skill diversity constitutes a complex research subject that requires a comprehensive analytic framework. A key methodological challenge in measuring skill diversity is accounting for the *relationships* between skills. The underlying rationale is simply that a skill profile is more diverse when it combines skills that are more dissimilar from one another. To account for skill-to-skill distances, we propose *Skill Embedding* as a relational framework, which maps a wide range of skills onto a multidimensional vector space, utilizing their co-occurrence patterns at the job level. Our approach is rooted in the concept that the proximity between two distinct skills can be deduced from their likelihood of co-occurring within a job and the similarity of their co-occurrence patterns with other skills in the labor market. As such, this skill-to-skill co-occurrence forms the foundation for an "embedding" that maps discrete skill categories to a vector space equipped with distance metrics, which can then be used to construct measures of skill-to-skill distances and diversity of skill profiles.[7]

Without accounting for the relationships between skills, measures of skill diversity can be misleading. Figure 3 illustrates this important point. In the left-most plot, we show the estimated normalized vectors of five broad skill groups, each represented by an arrow.[8] Arrows that point in similar directions indicate skills that are more closely related. To quantify, we will use cosine similarity as a metric, which we will describe in detail later. Here, engineering and information technology skills show up as two very similar skills, and personal care and service skills point toward a quite different direction from the above skills. Next, using these skill vectors, we demonstrate how considering the distances between skills can refine our skill diversity measure in the middle and right-most plots. Job A requires three closely related skills: (1) engineering, (2) information technology, and (3) business. Job B demands two distinct skills: (1) information technology and (2) marketing and public relations. If we only consider the number of skills required for each job, it might seem like Job A requires a broader skill set. However, when the average distances between these skills are taken into account, Job B exhibits greater skill diversity than Job A. Specifically, the skill profile in Job B has a higher average cosine distance (0.15 vs. 0.01) and a higher maximum cosine distance (0.15 vs. 0.02) than Job A.[9]

Overview of Analysis Pipeline

Our analysis pipeline begins by organizing skills extracted from online job postings into a skill-job matrix. Utilizing matrix decomposition techniques, we then project these skills onto a multidimensional vector space. This vector space functions as an embedding for the skills, where skill vectors with a higher frequency of co-occurrence are positioned closer

[7]While using skill co-occurrence to represent skills in a vector space has been proposed by recent studies (see Khaouja et al., 2021 for a review), our study is the first to use such a framework to measure skill diversity at the job level.

[8]For simplicity and without loss of generality, we project the estimated multi-dimensional skill vectors onto the two-dimensional plane in our illustration.

[9]For illustrative purposes, the example above groups skills into five broad categories, but our empirical analysis will employ a more detailed classification of skills. We return to this distinction in the presentation of our empirical findings.

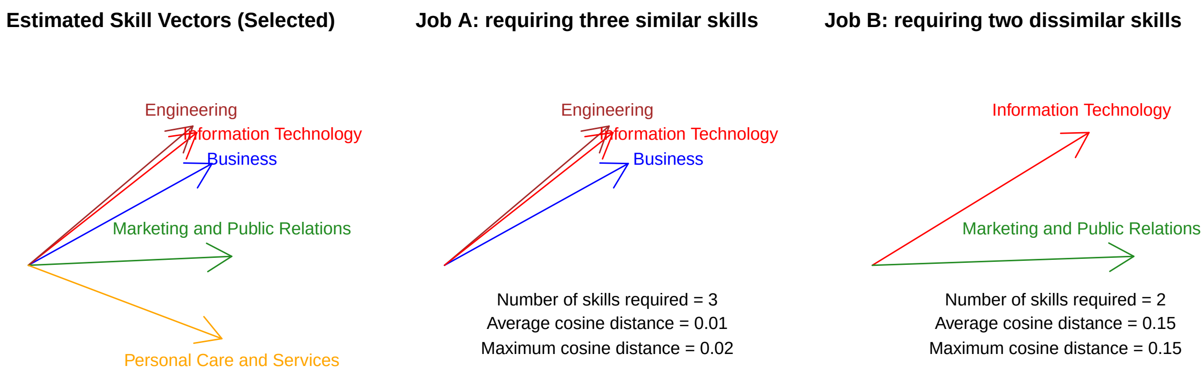


Figure 3: Why Skill-to-skill Relationships Matter: An Illustration of Two Hypothetical Jobs

Note: The skill vectors are based on empirical estimation from our data, which will be discussed in subsequent sections. Job A requires a greater number of skills than Job B, but Job B requires a more diverse set of skills.

Alt Text: Three plots of skill vectors as arrows from a common origin. Left: five broad skill groups shown in different directions. Middle: Job A with three closely clustered arrows (engineering, IT, business); average cosine distance 0.01. Right: Job B with two widely separated arrows (IT, marketing); average cosine distance 0.15.

together. The final step involves developing the measure of within-job skill diversity based on this skill embedding. Figure S1 in the Online Supplement presents an analysis pipeline to elucidate our methodological approach and empirical strategies. Part I of the Methodological Details section presents a mapping between conceptual and analytical elements of the Skill Embedding framework.

A Vector Representation of Skills

The first crucial step is to project skills into a multi-dimensional space to represent the relative distances between skill vectors. This projection starts with a m-skills-by-n-jobs **Skill-Job Matrix (denoted by A)**, which represents whether a specific skill is mentioned as a requirement in a job position:[10]

$$\begin{array}{cccccc} & \text{Job 1} & \text{Job 2} & \text{Job 3} & ... & \text{Job n} \\ \text{Skill 1} & 0 & 1 & 1 & ... & 0 \\ \text{Skill 2} & 1 & 0 & 0 & ... & 1 \\ \text{Skill 3} & 0 & 0 & 1 & ... & 1 \\ ... & ... & ... & ... & ... & ... \\ \text{Skill m} & 0 & 1 & 1 & ... & 0 \end{array} \tag{1}$$

The next step is to derive a vector representation of skills, so that skills that are closer in the vector space have a greater tendency to co-occur. This is done via Singular Value Decomposition (SVD).[11] We decompose the $m \times n$ skill-job matrix A into the product of three matrices:

$$A = V\Sigma U^T. \tag{2}$$

[10]This skill-job matrix contains information about the co-occurrence tendency between skills. To see this, note that the Skill-Skill Matrix (defined by $\mathbf{Q} = \mathbf{AA'}$) will be a symmetric ($Q' = (AA')' = AA' = Q$) with elements ($q_{ij}$) reflecting how often skill i co-occurs with skill j across all jobs ($q_{ij} = \Sigma_{k=1}^{K} p_{ik}p_{jk}$).

[11]The skill-job matrix to be used for SVD, by default, is centered and scaled.

The dimensions of V, Σ, and U^T are: $m \times m$, $m \times n$, and $n \times n$, respectively. Σ is the diagonal matrix of the singular values.

To preserve the relational structure of skills in the vector space while at the same time reducing noises in the data, we further employ SVD as a low-rank approximation technique, focusing on the first k singular values in Σ, and approximate matrix A using A_k: $A_k = V_k \Sigma_k U_k^T$, where the dimensions of V_k, Σ_k, and U_k^T are: $m \times k$, $k \times k$, and $k \times n$, respectively. In practice, the value of k can be determined by the elbow method, which is a graphical method to determine the important dimensions of the SVD. We will come back to this when we describe our empirical analysis. [12]

The decomposition outlined above serves two critical roles in our skill embedding framework. Firstly, it converts the skill-job matrix into a multi-dimensional vector space. Specifically, by taking the product of the row vectors of V_k with the matrix of singular values Σ, we obtain a vector representation of skills within this vector space. Each skill finds its representation through the i-th row of $V_k \Sigma_k$, denoted $\tilde{v}_i = (v_{i1}\sigma_1, v_{i2}\sigma_2, \ldots, v_{ik}\sigma_k)$. These skill vectors enable us to quantify the distances between different skills and explore the diversity of skill profiles within specific jobs or occupations. Secondly, our decomposition has a relational aspect. The skill embedding framework takes into account not only the *direct* associations between skills (e.g., skills A and B appearing in the same job) but also the *indirect* connections between skills through their co-occurrence patterns with other skills within the system (e.g., skills A and B often co-occurring with skill C). This feature extends beyond what conventional measures of skill co-occurrence, such as correlation and overlap, can encompass.[13] To illustrate this point, consider the example of analytical skill and coding skill. In our vector space, these two skills are not only in proximity due to their frequent co-occurrence in the same jobs but also because they share similar co-occurrence patterns

[12] The second panel of Figure S1 illustrates this SVD decomposition visually.

[13] In this sense, the skill embedding approach is methodologically analogous to word embedding models. Just as word-embedding models are not limited to first-order, direct semantic similarity but instead rely on the full network of word-to-word co-occurrence to infer both direct and indirect similarity, our skill embedding method likewise does not rely solely on first-order skill co-occurrence. Rather, it is driven by higher-order co-occurrence patterns across the entire skill network.

with other skills, such as data visualization and project management. Another helpful example is the relationship between caring for children and caring for the elderly. Although these two skills rarely co-occur in the same job, each is likely to co-occur with related skills, such as emotional support, communication, and basic health monitoring. Hence, their skill vectors will also be pointing towards similar directions in the vector space.[14] This relational perspective enables us to capture more nuanced associations between skills and accurately portray their interdependencies within the labor market.

Measuring Skill Diversity

Our measure of within-job diversity is based on the degree of dissimilarity between skill vectors. Using $< v_p, v_q >$ to denote the inner (dot) product of the two skill vectors, we compute the dissimilarity between a pair of skills p and q using the *cosine distance* between the two skill vectors ($d_{p,q}$), defined as 1 minus the cosine similarity between them:

$$d_{p,q} = 1 - Cos(\theta) = 1 - \frac{< v_p, v_q >}{\|v_p\| \cdot \|v_q\|}, \tag{3}$$

As we saw earlier in Figure 3, the cosine distance measure directly corresponds to the angle formed between the skill vectors. A wider angle between skill vectors indicates a greater cosine distance and, consequently, a higher level of skill diversity. Conversely, a narrower angle signifies a lower cosine distance and a lower level of skill diversity. As expected, the figure reveals a close relationship (i.e., small cosine distance) between engineering and information technology skills, as their skill vectors point in similar directions, whereas personal care and service skills are noticeably distinct (i.e., large cosine distance) from the other skills.

[14]On the other hand, we do differentiate between similar skills based on the contexts in which they are applied. In the EBG job postings data, child care is classified under the "child care" skill category, while nursing home care falls under "medical skills." Although both involve caregiving, they differ in focus and context: child care centers on supporting children's development in educational or home settings, whereas nursing home care involves assisting elderly individuals with health-related needs. As such, they are not inherently the same type of skills. Therefore, their skill vectors will point to similar, but not the same direction.

This baseline skill dissimilarity measure can then be used to compute within-job skill diversity. Consider job j, which has a total of M required skills. We propose two parallel measures to capture the concept of *within-job skill diversity* for this job. First, we measure it as the *average pairwise cosine distance* across all skill vectors within job j:

$$\text{Average Pairwise Cosine Distance} = \delta_j^{within} = \frac{2}{M(M-1)} \Sigma_{p,q \in \Theta_j \& p<q} d_{p,q}, \tag{4}$$

where the constant factor in front of the summation sign is the inverse of the total number of possible skill pairs ($\frac{M(M-1)}{2}$), and Θ_j denotes the set of skill vectors in job j. We can also calculate occupation-level within-job skill diversity as the mean of δ_j^{within} across all jobs in this occupation.[15]

The average cosine distance, as defined above, can be interpreted as the typical level of dissimilarity between a *randomly selected pair of skills required* for a given job. Alternatively, within-job skill diversity can also be measured by the *maximum pairwise cosine distance* among all possible skill pairs in a job, which is expressed as:

$$\text{Maximum Pairwise Cosine Distance} = \max_{p,q \in \Theta_j \& p<q} d_{p,q}. \tag{5}$$

Conceptually, these two alternative measures capture distinct dimensions of within-job skill profiles: one reflects the average dissimilarity across all skill pairs, while the other captures the maximum span of required skills. Substantively, however, our empirical analysis (presented later) shows that these measures are highly correlated and yield consistent patterns and trends in skill diversity. Notably, their correlation with each other is substantially stronger than with the simple count of required skills, underscoring the importance of incorporating skill dissimilarity into measures of skill diversity. For future research using our methodology, we offer two recommendations. First, as a general practice, we suggest conducting robustness checks using both the average and maximum distance measures. Sec-

[15]For occupations with only one skill required, the skill diversity measure will be zero.

ond, the average distance measure is generally less sensitive to outliers and more stable and comparable across samples, as it reflects the full distribution of pairwise dissimilarities rather than relying on a single extreme value (i.e., the maximum). Therefore, when using the maximum distance, we recommend an additional robustness check that substitutes the average of the two highest cosine distances for the single maximum value. This approach helps mitigate potential distortions caused by outliers.

For readers interested in more details of our skill diversity measure, we recommend referring to the Methodological Details section at the end of the main text for a summary of the conceptual and analytical elements of the skill embedding framework (Part I) and numeric examples of the difference between within-job and between-job skill diversity (Part II).

Data and Measures

Capturing Skill Diversity From Job Postings

We use Emsi Burning Glass (EBG, now Lightcast) job postings data to construct an embedding of fine-grained skills in the labor market and compute skill diversity measures. The EBG data set is gathered by scraping over 45,000 websites worldwide, including company career sites, national and local job boards, and job posting aggregators. The scraping process follows a fixed schedule of browsing job websites that are carefully monitored and updated to include the most current and complete set of online postings. Our sample focuses on ten years of EBG job postings data from 2011 to 2020 in the United States. Each year, the data are compiled by month as originally produced by the data provider. This enables us to obtain job postings from 120 time points (10 years X 12 months). After deduplicating the job postings, the company further extracted information such as job title, company, skills, and educational requirements from each job posting. It is estimated that online job ads capture about 85% of the total job openings in the labor market (Lancaster et al., 2019). Recent labor market studies have increasingly utilized the EBG data to analyze various skill dimensions (Deming and Kahn, 2018; Hershbein and Kahn, 2018; Wilmers and Zhang, 2022), but to our knowledge, no prior work has used this data to study skill diversity. Because of the vast number of job postings contained within this dataset, performing matrix decomposition and computing pairwise cosine distance on the entire dataset of job postings would result in an excessively long computational time. To reduce computational burden, we randomly selected a 0.1% sample from each calendar year, yielding approximately 15,000 to 36,000 job postings annually. This corresponds to an overall average of 26,309 job postings per year and a total of 263,088 job postings over the 10-year period—a scale substantially larger than that of conventional survey-based analyses. Table S1 in the Online Supplement reports the sample sizes by year and month.

Choosing job postings data as our primary data set for constructing the skill embedding framework holds distinct advantages over various other potential data sources, making it particularly suitable for our study's objectives. Firstly, the skill information garnered from job postings presents a unique level of granularity and specificity. Unlike more commonly utilized datasets such as O*NET, our dataset derived from the EBG provides insights into skill requirements at the job level, nested within occupations.[16] As discussed earlier, this granularity enables us to separate out within-job skill diversity from between-job skill diversity. Secondly, the job postings data provides detailed information on a rich array of skill requirements. Employers leverage these postings to attract candidates who are not just qualified, but also well-matched to the specific demands of the position. Thus, the skills highlighted in these postings can be interpreted by employers as the most critical for success in a given role.[17] In contrast, currently available sources of job requirements and duties of workers themselves are much shorter—for example, the mean number of words for verba-

[16]Recently, Tong et al., 2025 also utilized the EBG data to analyze skill upgrading over time, but their analysis focused on the occupation-level skill profiles rather than at the job level.

[17]Empirical validation of this assumption is found in recent studies which demonstrate that, after accounting for education and experience, the skill requirements advertised in job postings are indicative of wage variations across firms and local labor markets, even within narrowly defined occupations (Deming and Kahn, 2018; Hershbein and Kahn, 2018).

tim task descriptions given by GSS respondents is 12.11 (Martin-Caughey, 2021). Thirdly, unlike cross-sectional labor force surveys, which include existing employment relations over varying durations, job postings offer a real-time stream of data on new job opportunities as they become available. This results in a data set that is more attuned to the current and evolving demands of the labor market. These features make the EBG data a unique resource for constructing the skill embedding framework in our study.

Despite these advantages, job postings data present some important limitations. The first limitation comes from its lack of representativeness of the occupational composition of the labor force (Lancaster et al., 2019). For example, the data may fail to capture job postings in offline sources (e.g., print newspapers, television, or radio ads) or via word-of-mouth. In addition, a single online ad might represent multiple openings with the same job description, which means that the EBG data will undercount the total number of job openings, and this undercount may differ by industry or occupation (Modestino et al., 2026). Indeed, by our calculation of data from the 2010s, the EBG data overrepresented sales and management/professional occupations (see Figure S7 in the Online Supplement).[18] We adopt two strategies to mitigate this limitation. First, for analyzing patterns and trends, we integrate skill diversity measures derived from the EBG data with large-scale, nationally representative data from the American Community Survey. Second, recognizing that the non-representativeness of EBG data may distort the skill-job matrix used to generate the skill embedding space, we conducted robustness checks using reweighting techniques to adjust the matrix to better reflect the population distribution. Specifically, when estimating the skill embeddings, we weight the skill-job matrix by applying weights to job postings that adjust for the over- or under-representation of detailed occupations in a given year.[19] The resulting embeddings yield consistent findings, as reported in Figure S10 in the Online Supplement.[20]

[18]Other studies also pointed out that EBG data underrepresent job vacancies in the agricultural and construction sector (Hershbein and Hollenbeck, 2014).

[19]The weights are proportional to the occupational proportion in the EBG data divided by the occupational proportion in the ACS data.

[20]Despite these efforts, we note that even after adjusting for selection by (detailed) occupational composition, our data cannot account for job-level selection into online posting *within* detailed occupations. Moreover,

The second limitation is that, by relying on job postings, we are constrained to the types of skills and descriptions employers choose to include. Some skills may be grouped together when further specification is unnecessary, while others may be omitted entirely—either because they are basic and assumed competencies or because they are subtle and rarely acknowledged in formal contexts[21]. For instance, although writing skills are essential for a sociology professor, a posting for such a position is unlikely to list "writing" explicitly, as it is taken for granted as a baseline qualification. Instead of covering all skills utilized, job postings tend to emphasize skills that signal what the employer values most for the role, attract potential applicants, and help identify candidates who meet specific requirements. Moreover, these requirements more accurately reflect the employer's perspective, not the employee's; they may not capture the skills workers actually use or perceive as essential in their day-to-day work.[22] Despite these limitations, the detail and coverage of job postings—down to the job level—provide a uniquely rich source for investigating skill diversity on a large scale.

Extracting Skill Requirements

Skills were extracted from these job postings through a systematic, multi-step process. First, over 10,000 unique keywords and phrases identified by EBG were distilled into a subset of general "job skills" relevant across a wide range of occupations. EBG then cleans and codes the text of job ads into a taxonomy of thousands of unique but standardized requirements. Starting from a predefined list of possible skills, EBG employs machine-learning algorithms to scan each posting for indications that a skill is required. These extracted skills are then

such selection is likely to vary over time, reflecting changes in various factors such as HR practices, firm-specific hiring strategies, the balance between internal and external labor markets, and local labor supply conditions.

[21] The skill or "in-betweenness" among second-generation immigrants—that is, the skill to navigate between cultural and linguistic worlds in a workplace that is otherwise highly segregated along racial, class, and generational lines—discussed in Wilson, 2018 offers an example of informal, subtle skills in the workplace that's rarely formally recognized in job postings.

[22] It is also possible that the magnitude and nature of these biases vary over time. For example, as online job postings have become more widespread, employers may increasingly include longer wish lists of desired skills in their postings.

aggregated into skill clusters—groups of skills that tend to co-occur in education or training programs or are substitutable across labor market contexts. The clusters are generated using a combination of statistical clustering methods and expert curation. In the EBG data, these clusters are further organized into cluster families, which map roughly onto broad career areas such as Information Technology, Finance, and Health Care. As such, the final EBG dataset contains measures of skill requirements on three levels, ranging from the most detailed one (about 13,500 categories), to the moderately detailed one (about 650 categories), to the broadest one (about 30 categories). We use the moderately detailed skill categorization, consisting of 535 skills consistently appearing across the ten years. This allows us to obtain reasonably fine-grained groupings of skills while at the same time reducing the sparseness of skill-job matrices. Figure A2 in Part IV of the Methodological Details sections presents examples of skill categories in the EBG data for data science skills (upper panel) and childcare skills (lower panel).

EBG coded each job posting into the 2010 Standard Occupational Classification (SOC) scheme, in alignment with the occupational categorization scheme available in the American Community Survey data to be used in the next stage of analysis. In the EBG dataset, skills are originally represented in a series of numbered columns. Most jobs do not report more than ten skills (see Figure S4 for the distribution of the number of skills mentioned in each job). As a result, the skill-job matrix exhibits significant sparsity. We first combine all skills across the above-mentioned skill columns into a single textual paragraph. We then use a count vectorizer to identify and tally the occurrences of each skill in a given job. The dictionary used for this count vectorization comprises the total number of unique skills present in the operational sample. In this way, we obtained matrices on the scale of 535 skills-by-26,308 jobs, capturing the skill profiles of different jobs as well as the co-occurrence patterns between different skills each year.[23] Part III of the Methodological Details section presents some examples of original job postings with varying degrees of skill diversity.

[23]Figure S5 in the Online Supplement reports the number of skills not included in the common set of 535 skills by year. We find no systematic pattern over time.

Constructing Skill Diversity Measures and Merging with ACS Data

Utilizing the skill-by-job matrix, we conduct the Singular Value Decomposition (SVD), which further decomposes the matrix into a vector space comprising 535 skills, as detailed in the methods section. Not all skill dimensions contributed equally to the overall variance. To see this, Figure S6 shows a scree plot with the sorted singular values for each skill dimension, ranging from highest to lowest contribution to overall variation. Using the elbow method described earlier, we identify the top 20 values as providing an adequate representation of the differences in skill requirements across jobs.[24] To reduce dimensions, we retain the top 20 dimensions of skill vectors, which accounted for the majority of the total variation in skill-to-skill co-occurrence while reducing the noise in the data. We then follow the steps outlined in Equations 4 and 7 to compute measures of within-job skill diversity.

We merge the year-specific skill diversity measures at the detailed occupational level with the American Community Survey (ACS) data from 2010 to 2017.[25] To ensure consistency in occupational coding, we rely on the occupational coding scheme measured consistently across both the EBG and ACS data, namely the SOC 2010.[26] Throughout the

[24] Here, the diagonal values of Σ make up the singular value spectrum, visualized in a one-dimensional plot referred to as the "scree plot" (Cattell, 1966). Each singular value indicates its importance in explaining the data. Specifically, the square of each singular value is proportional to the variance explained by each singular vector. If the original variables are linear combinations of a smaller number of underlying dimensions, combined with some low-level noise, the plot will tend to drop sharply for the singular values associated with the underlying dimensions and then much more slowly for the remaining singular values. In the scree plot, singular values to the right of such an "elbow" are ignored because they are assumed to be mainly due to noise.

[25] Since the merge relies on year-occupation level, the ACS-based skill diversity trends derived from our approach capture two key sources of change. First, because the merge is done with year-specific estimates, the trend will reflect temporal changes in within-job skill diversity for a given detailed occupation. For example, if lawyers (a detailed occupation) are increasingly required to possess a more diverse set of skills, this will be reflected as an upward trend in skill diversity among lawyers in both the EBG data and ACS data. Second, the trends also reflect compositional changes across detailed occupations within a given occupational group, such as the disproportional growth of occupations with higher or lower average skill diversity. For instance, within professional occupations, if the share of chemical engineers (who exhibit lower within-job skill diversity by our estimates) declines while the share of aerospace engineers (who exhibit higher within-job skill diversity) rises, the overall skill diversity within the professional occupational group will increase. Robustness checks using the skill embedding estimated from 10-year combined EBG data yielded consistent findings (see Figure S11 in the Online Supplement).

[26] We exclude years from 2018 onwards to maximize the coherence in occupational coding, as the ACS transitioned from using the 2010 SOC scheme to the 2018 scheme starting from 2018.

years included in the analysis, the merging rate between the skill diversity measures and the ACS data ranges from approximately 90% to 93%. The American Community Survey is an extensive and nationally representative survey conducted by the United States Census Bureau, systematically collecting data on various demographic, social, economic, and housing characteristics of the U.S. population on an annual basis. As a result, when compared to the EBG dataset, which may be subject to sample selectivity in the types of occupations in job postings (Lancaster et al., 2019), the ACS sample offers a more accurate representation of labor market trends within the broader population. Figure S7 compares the occupational composition between EBG and ACS data. We focus specifically on prime-age individuals aged between 25 and 64 who have a primary occupation.[27] For the subsequent analyses, we categorize detailed occupations into distinct aggregate groups: management, professional, healthcare support, personal care and service, other service, sales, office and administrative, construction and production.[28] This grouping allows us to explore variations between different types of occupations. In all the ACS-based analyses, we employ ACS person weights to ensure the results are representative of the national distribution.

Empirical Results

Validation #1—the Importance of Considering Skill Relationships

Before presenting the empirical trends in skill diversity, we first assess the validity of our measures in two important ways. The first is to evaluate whether the Skill Embedding framework offers value beyond existing measures. Our analytical framework posits that to accurately measure skill diversity, one must consider the *varying* levels of similarity between different skills. To demonstrate this empirically, we take a closer look at the ten most commonly listed skills in job postings across three occupational categories in Figure 4: managers,

[27] For workers employed during the previous week, the variable refers to the job at which the person worked the greatest number of hours. For persons who were unemployed or out of the labor force, the variable refers to the most recent job, if it was within the previous five years.

[28] We exclude military occupations from the sample.

engineers, and personal care and service workers. In managerial jobs, for instance, the skill of "budget management" has a high degree of overlap with "general accounting" due to their shared focus on fiscal resource allocation. However, it aligns far less with "operations management," which is more concerned with the daily processes of producing goods or services. This lack of similarity isn't unexpected when considering the distinct objectives of each skill dimension. Within engineering jobs, certain skills demonstrate a strong similarity (e.g., "civil and architectural engineering" and "drafting and engineering design"), whereas others show relatively little overlap (e.g., project management versus equipment repair and maintenance). In personal care and service jobs, we see that "basic customer service" skills are closely connected to "general administrative and clerical tasks," but show a much weaker connection to tasks like "housekeeping" or "child care." These observed patterns underscore that the degree of skill similarity can vary significantly, even among highly similar jobs. Recognizing and incorporating these skill relationships is essential for a nuanced understanding of skill diversity within and between jobs.

Another way to highlight the importance of accounting for varying degrees of skill-to-skill similarity is to compare our cosine-distance-based measures with a traditional metric: the raw count of skills required in a job. Figure 5 presents pairwise scatterplots and correlations among three measures of skill diversity: number of skills, average cosine distance, and maximum cosine distance. The latter two are derived from our estimated skill embedding, which reflects the relationships between skills, and they are highly correlated ($\rho = 0.97$). In contrast, both show substantially lower correlations with the number of skills ($\rho = 0.69$ and 0.59, respectively), a measure that ignores differences in distances across skill pairs. These results suggest that the relational, embedding-based measures capture dimensions of skill diversity overlooked by simple skill counts.

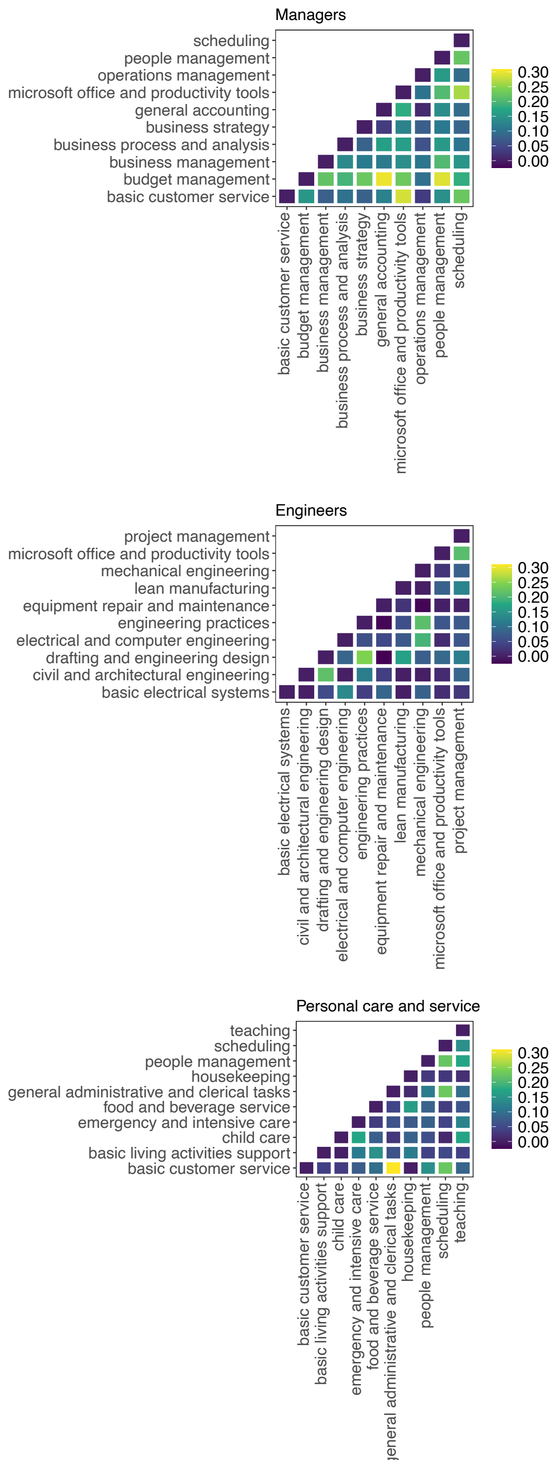


Figure 4: Estimated Cosine Similarity among the Ten Most Common Skills in Managers, Engineers, and Personal Care and Service Occupations

Alt Text: Three triangular heatmaps showing cosine similarity among the ten most common skills in managerial, engineering, and personal care and service jobs. Color intensity ranges from 0 (dark) to 0.30 (yellow). Related skill pairs within each occupation appear as brighter cells; unrelated pairs appear darker.

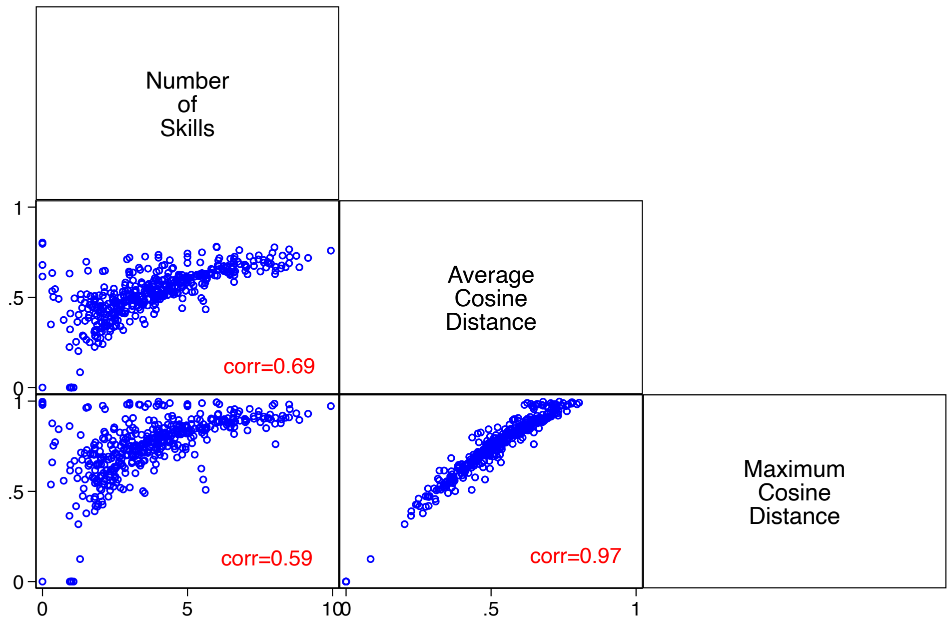


Figure 5: Pairwise Associations among Three Measures of Within-job Skill Diversity

Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017.

Alt Text: A scatterplot matrix comparing number of skills, average cosine distance, and maximum cosine distance across jobs. The correlation between average and maximum cosine distance is 0.97. Both cosine measures correlate more weakly with number of skills: 0.69 for average and 0.59 for maximum.

Validation #2—the Importance of Job-level Analysis

Our second validation focuses on the unique contribution of job-level analysis. As illustrated earlier in Figure 2, within- and between-job skill diversity capture distinct dimensions of occupational skill profiles. Do these distinctions appear in real-world data? Figure 6 confirms that they do. Here, we present the distributions of within-job (top left) and between-job (top right) skill diversity across occupational groups, along with their relationship at the detailed occupation level (bottom), where circle sizes reflect occupation size. The figure shows that both measures exhibit substantial variation across and within occupational groups but are only weakly correlated (correlation=0.05). While prior occupation-level analyses conflate these dimensions, our job-level approach distinguishes them explicitly.

Main Findings: Levels and Trends in Skill Diversity across Occupational Groups

Figure 7 shows estimated trends in within-job skill diversity across occupational groups. Bold blue lines highlight three focal categories—management, professional, and personal care and service—which exhibit the greatest employment expansion over this period. To examine these patterns more closely, Table 1 lists occupations with the highest and lowest skill diversity within each of these groups. As shown in the left panel of Figure 7, management occupations have the highest level of skill diversity, with a clear upward trend. Professional occupations also show relatively high diversity, though their growth is more modest. Personal care and service jobs began with the lowest level of within-job skill diversity but experienced the steepest increase over the study period.

To better compare these trends, the right panel of Figure 7 plots changes in within-job skill diversity by occupational group over the past decade. Values are normalized to zero in the baseline year (2010), with the Y-axis representing the percentage increase relative to that baseline. Within-job skill diversity increased by approximately 30% during the 2010s for personal care and service jobs, compared to a more modest rise of about 10% for management jobs and just 3% for professional jobs. That is, in percentage terms, the increase in skill

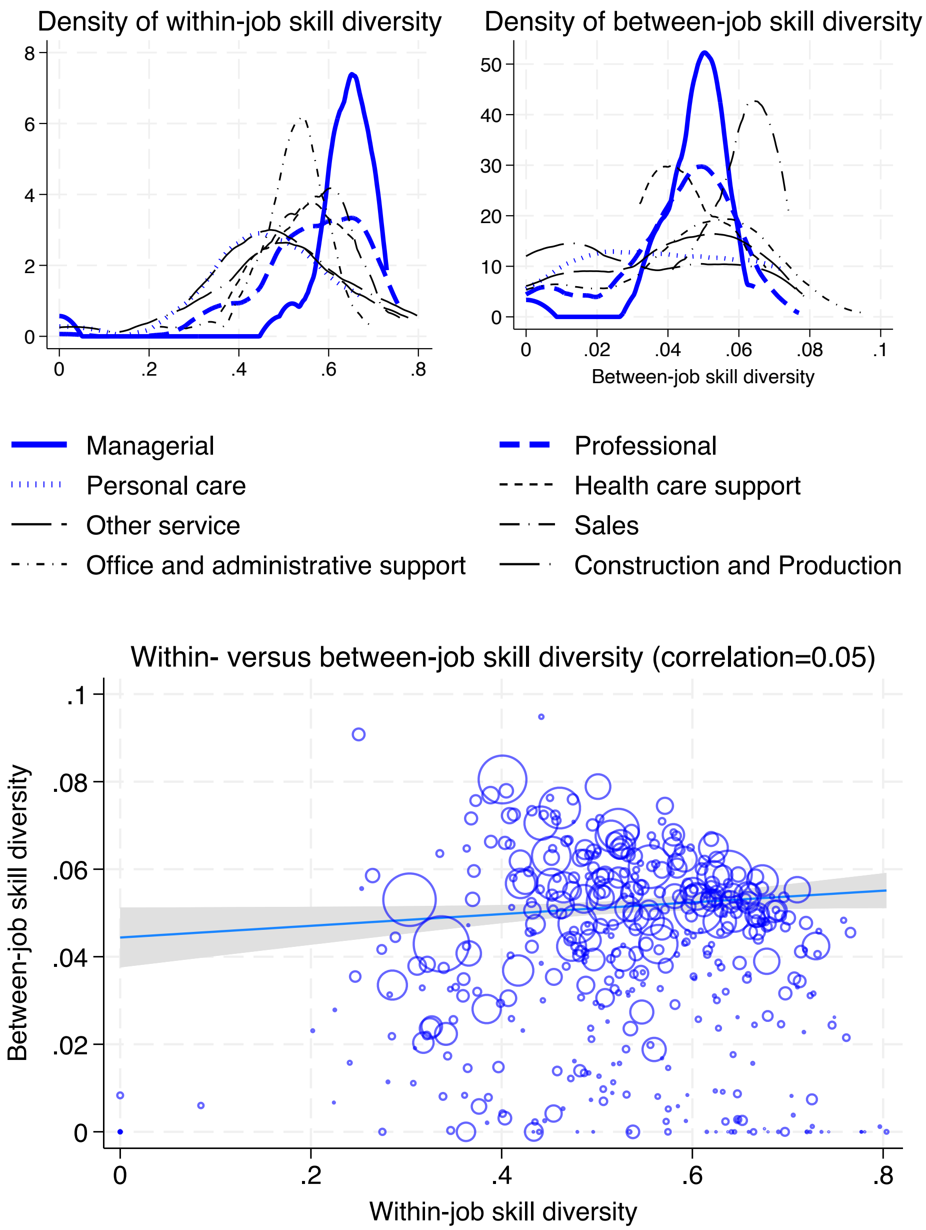


Figure 6: Distributions of Within-job and Between-job Skill Diversity Measures and Their Relationship

Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017.

Alt Text: Three panels. Top left: density distributions of within-job skill diversity across eight occupational groups, ranging roughly 0 to 0.8. Top right: density distributions of between-job skill diversity, ranging roughly 0 to 0.1. Bottom: scatterplot of within-job against between-job skill diversity at the detailed occupation level; correlation 0.05.

diversity among low-paying service jobs is three times as large as the corresponding trend in management occupations. These patterns indicate that the rise of skill diversity has not only extended beyond high-paying occupations but has been most pronounced in the lowest-paying group — personal care and service.

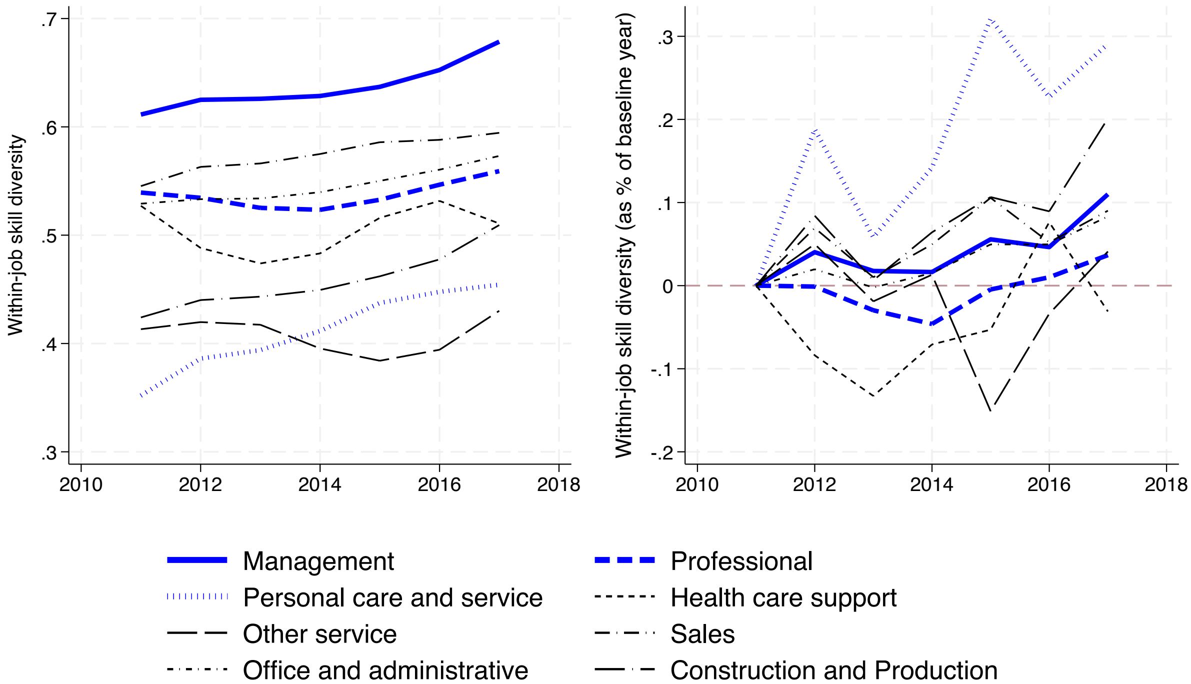


Figure 7: Trends in Within-job Skill Diversity: Raw Values and Percentage Change Relative to 2010

Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017.

Alt Text: Two line charts across eight occupational groups. Left panel: raw values, 0.3 to 0.7; management highest, personal care and service lowest but rising steepest. Right panel: percentage change from 2010; personal care and service rises to about 30 percent, management about 10 percent, professional about 3 percent.

Would the same trend emerge if we measured skill diversity simply by the number of skills, ignoring relationships between them? Our results suggest otherwise. Figure 8 compares two alternative measures: number of skills (left panel) and maximum cosine distance (right panel). Based on the number of skills alone, the three focal occupational groups ap-

Table 1: Detailed Occupations with the Lowest and Highest Within-job Skill Diversity within Management, Professional, and Personal Care and Service Occupations

Occupation	Mean Cosine Distance
Management	
Lowest five diversity	
Farmers, Ranchers, and Other Agricultural Managers	0.50
Food Service Managers	0.54
Property, Real Estate, and Community Association Managers	0.58
Lodging Managers	0.59
Chief Executives	0.61
Highest five diversity	
Computer and Information Systems Managers	0.69
Architectural and Engineering Managers	0.69
Administrative Services Managers	0.70
Purchasing Managers	0.71
Marketing Managers	0.73
Professional	
Lowest five diversity	
Tax Preparers	0.27
Optometrists	0.32
Dietetic Technicians	0.33
Elementary School Teachers, Except Special Education	0.34
Dentists, General	0.34
Highest five diversity	
Chemical Engineers	0.73
Environmental Engineers	0.73
Chemical Technicians	0.75
Agricultural Engineers	0.75
Aerospace Engineers	0.77
Personal Care and Service	
Lowest five diversity	
Childcare Workers	0.29
Baggage Porters and Bellhops	0.34
Barbers	0.35
Gaming Dealers	0.37
Nonfarm Animal Caretakers	0.41
Highest five diversity	
Residential Advisors	0.55
Tour Guides and Escorts	0.61
Gaming Supervisors	0.63
Ushers, Lobby Attendants, and Ticket Takers	0.66
Motion Picture Projectionists	0.68

pear to show similar increases in skill diversity. However, both the average and maximum cosine distance measures reveal a substantially greater rise in within-job skill diversity among personal care and service jobs. This suggests that while all three groups experienced comparable growth in the *quantity* of listed skills, it is the personal care and service jobs that saw the largest increase in the *dissimilarity* among those skills. Thus, only by accounting for relationships between skills can we detect the distinctive pattern of diversification occurring in lower-paying personal service occupations.

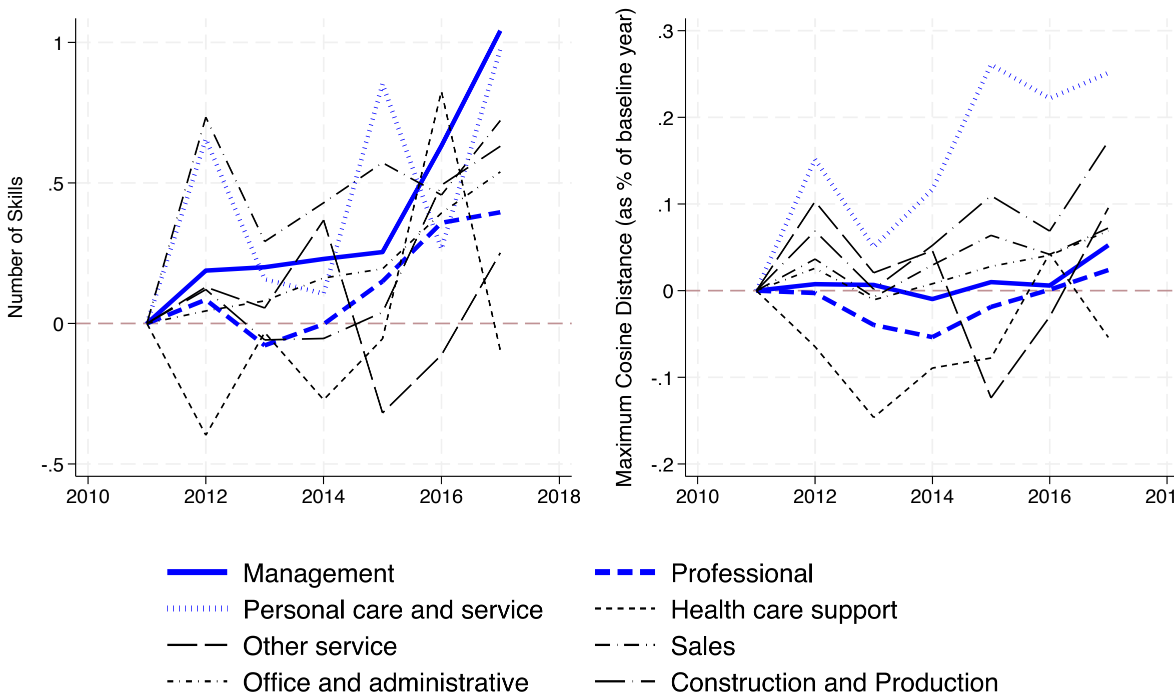


Figure 8: Trends in Within-job Skill Diversity Using Two Alternative Measures: Number of Skills and Maximum Cosine Distance

Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017.

Alt Text: Two line charts across eight occupational groups. Left panel plots standardized number of skills; all groups trend upward with similar magnitudes. Right panel plots maximum cosine distance in percentage terms; personal care and service rises most steeply, while professional and other groups show smaller increases.

How unique is the pattern of personal service and care jobs among all service occupations? Figure 9 presents the trends in skill diversification for different types of service jobs. The figure shows that the increase in within-job skill diversity in personal care and service jobs is clearly a distinct pattern not seen in other service occupations. It is this lowest-paying, rapidly expanding group of service occupations that has been driving the growth of skill diversity in low-wage jobs.[29]

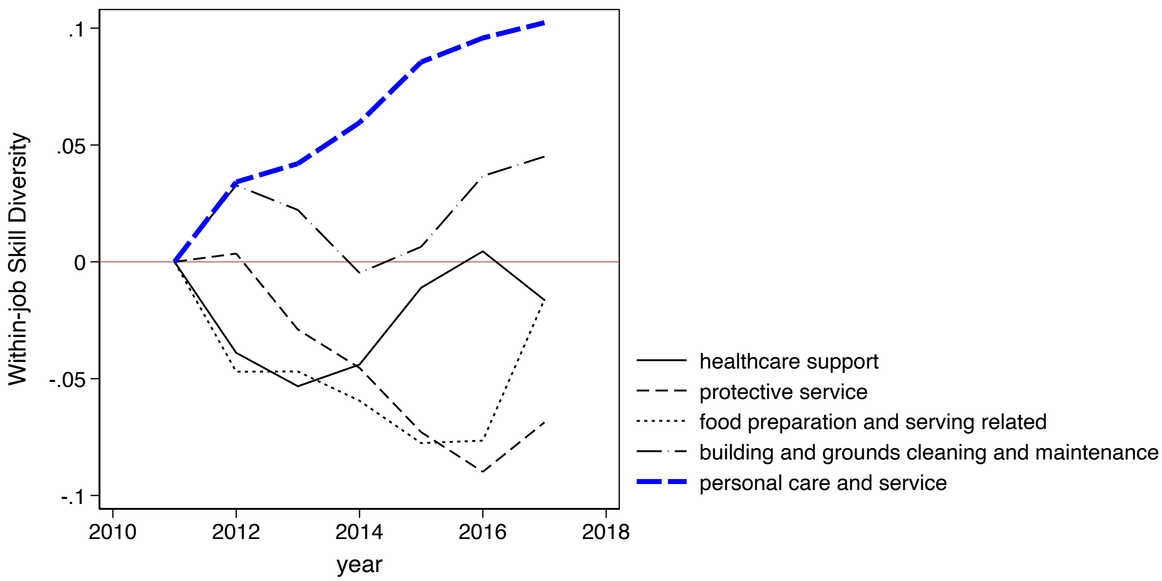


Figure 9: Trends in Within-job Skill Diversity across Service Occupational Groups

Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017.

Alt Text: A line chart, 2010-2018, of within-job skill diversity for five service occupational groups: healthcare support, protective service, food preparation, building maintenance, and personal care and service. Personal care and service (thick blue) rises steadily to about 0.10, while the other four groups remain flat or decline slightly.

Finally, we assess whether the observed trends are driven by changes within particular industries or states. Figure 10 displays within-job skill diversity trends across two-digit

[29]For readers interested in disaggregated groups within professional occupations, please see Figure S9 in Online Supplement.

industries and individual states (gray lines), with the red line in each panel indicating the overall average. The results suggest that the key pattern observed above—a sharper rise in skill diversity among personal care and service jobs compared to management or professional jobs—is broadly consistent across both industries and states.

Toward Future Research: Two Proposed Extensions

The primary aim of our empirical analysis is to examine trends in skill diversity across occupational groups. We find that skill diversification has occurred in both high- and low-paying occupations—a notable finding that draws attention to jobs often labeled as "low-skilled" and highlights the evolving, increasingly diverse skill requirements of these jobs. Looking ahead, two extensions can situate our findings within broader discussions of labor market transformation and inequality. First, it is important to consider the potentially heterogeneous experiences of workers in these occupations. Second, to link these trends to earnings inequality, future work should examine the returns to skill diversity across occupational groups. While each of these directions warrants a dedicated, in-depth analysis beyond the scope of this study, we offer preliminary findings on both fronts and outline promising avenues for future research.

Extension #1: Uneven Trends by Education and Income

The trends in skill diversity documented by our main results may mask significant differences among subgroups *within* occupational groups. To explore these variations, Figure 11 presents trends in within-job skill diversity by individuals' education and income levels (with tercile 1 representing the lowest and tercile 3 the highest). In management occupations, the increase in skill diversity varies moderately across education and income groups: individuals with higher education or income levels initially experienced greater increases in skill diversity, though these differences largely converged by 2017. In contrast, professional and personal care and service occupations show a different pattern — skill diversification appears more

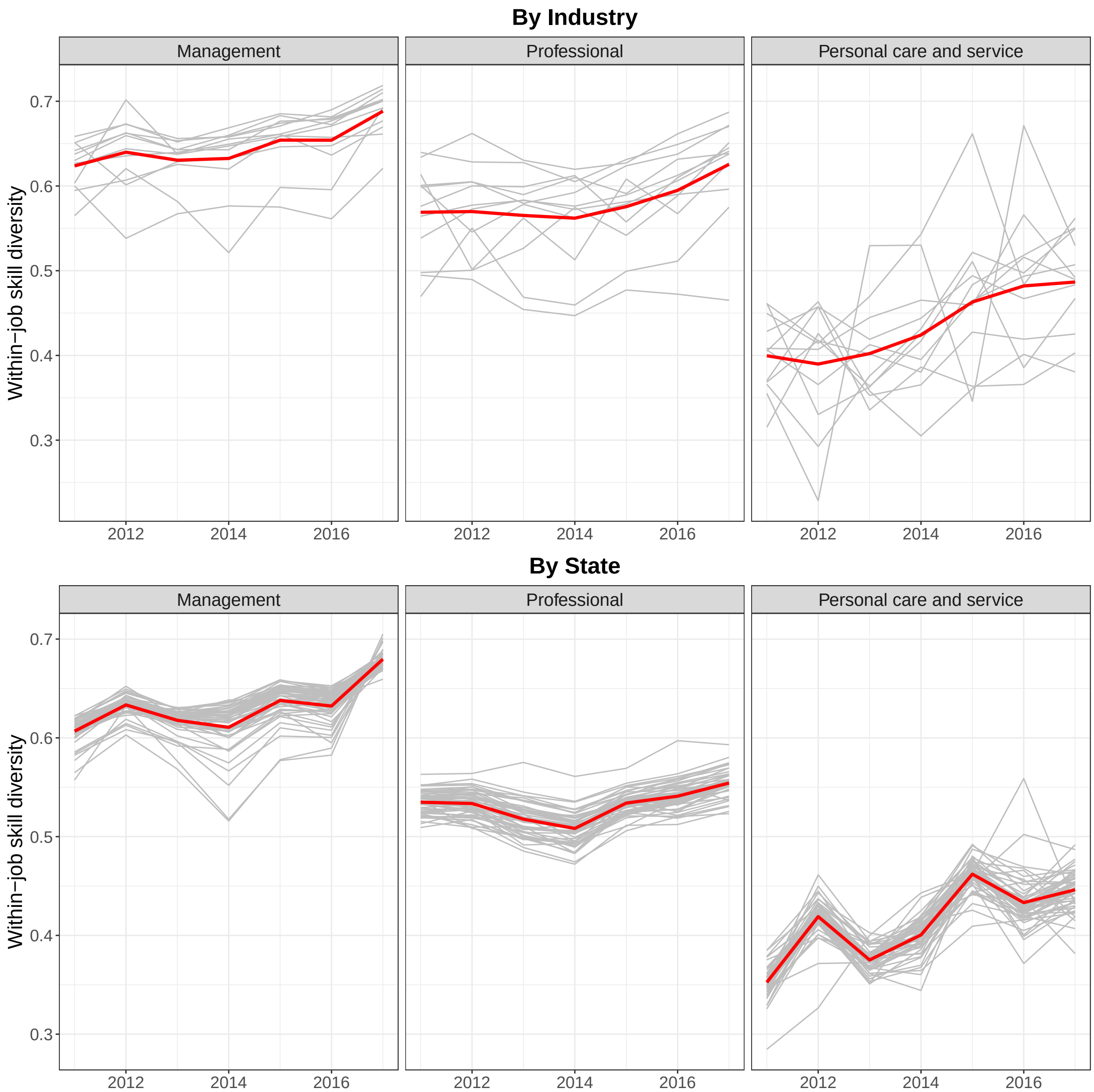


Figure 10: Trends in Within-job Skill Diversity for Three Focal Occupational Groups, by Industry and State

Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017.

Alt Text: Six panels arranged in two rows. Row A (by industry) and Row B (by state) each show three panels for management, professional, and personal care and service. Gray lines represent individual industries or states; red lines show overall averages. Personal care and service trends upward in both rows.

pronounced among workers without a college degree or with lower income levels. These findings suggest that skill diversification across occupational categories may be driven by distinct subgroups of workers. Notably, while skill diversification in management occupations is driven primarily by highly-paid workers, it is those at the *lower end* of the education and income spectrum who are increasingly combining a broader range of skills in professional and personal care and service jobs. Put simply, skill diversification is not only more pronounced in low-paying occupational groups, but also more concentrated in the lower-paying jobs *within* those groups. This trend may reflect either vertical upgrading—lower-status workers incorporating more advanced or technical skills—or horizontal enrichment, involving the addition of skills not previously typical for these roles. Additionally, the diversification of skills among low-education, low-income workers may also stem from declining bargaining power and rising job insecurity, which compel workers to demonstrate greater versatility in order to secure and retain employment. Exploring the mechanisms and specific skill dimensions underlying these subgroup variations presents a promising direction for future research.

Extension #2: Unequal Earnings Returns to Skill Diversity

Does within-job skill diversity pay? While the main goal of this study focuses on the uneven trends in within-job skill diversity, examining its earnings returns would further help us understand the inequality consequences of skill diversification. To achieve this, we estimate regression models using the individual-level ACS data predicting log annual income (Y_i):

$$Y_i = \beta_0 + \underbrace{\gamma_w \tilde{\delta}^{within} + \gamma_b \tilde{\delta}^{between}}_{\text{skill diversity}} + \underbrace{\Sigma_{m=1}^{20} \kappa_m S_{im}}_{\text{mean skill vector}} + \underbrace{\Sigma_{k=1}^{K} \beta_k X_{ik}}_{\text{controls}} + \epsilon_i, \tag{6}$$

where $\tilde{\delta}^{within}$ and $\tilde{\delta}^{between}$ represent normalized measures of within-job and between-job skill diversity for an individual's primary occupation, respectively. The calculation of between-job skill diversity is described in Part II of the Methodological Details section. To

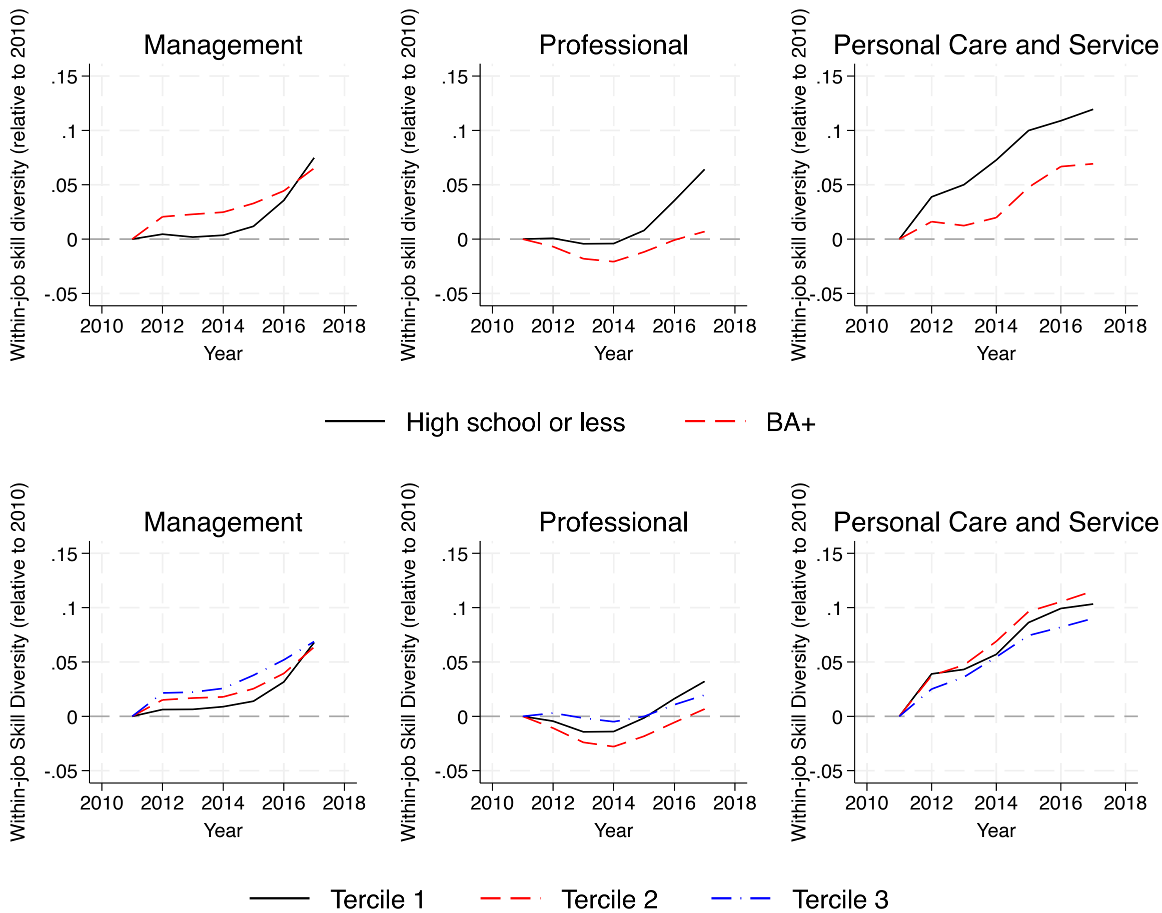


Figure 11: Trends in Within-job Skill Diversity by Individual Education and Income

Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017.

Alt Text: Six panels across education (Panel A) and income (Panel B), each showing management, professional, and personal care and service. Panel A compares high school or less against BA-plus workers. Panel B compares income terciles. Low-education and low-income workers drive skill diversification in professional and personal care jobs.

facilitate interpretation, we normalized skill diversity measures to have a mean of zero and a standard deviation of one. As jobs characterized by greater skill diversity might also demand unique, specialized skills that are positively associated with earnings, we isolate the impact of skill diversity from that of the specific skill dimensions by controlling for the top 20 most important skill dimensions in the average skill vector of the occupation. Here, the importance of a skill dimension is determined based on its relative contribution to the overall variance, as detailed in the SVD analysis in the methods section. The models also control for individual-level covariates (X_{ik}). These include year dummies, gender, education level (less than high school, high school degree, associate degree, college degree, and advanced degree), Duncan SEI index of the primary occupation, race, and Hispanic origin. The models are weighted to ensure representation of the general population.[30]

Figure 12 shows the coefficients reflecting the earnings effects of standardized skill diversity measures. The results for both mean cosine distance and maximum cosine distance measures indicate a consistent and statistically significant positive link between within-job skill diversity and earnings across the board. These effects are significant even after accounting for the impact of specific skill dimensions. However, the magnitude of earnings gains associated with 1 SD increase in within-job skill diversity varies substantially across occupational groups: the coefficient on mean cosine distance ranges from about 2% (construction and production) to about 22% (sales). Importantly, the three rapidly expanding occupational groups witnessed *unequal* economic returns to within-job skill diversity: using mean cosine distance as the skill diversity measure, a 1 SD increase in within-job skill diversity is associated with a 15% earnings premium for management occupations, but this earnings premium drops to 12% for professional occupations and further drops to 3.5% in personal care and service occupations. Hence, working in job positions that require a wide range of skills is particularly beneficial for management occupations. This may reflect a combination

[30] Preliminary analysis included year-by-year earnings regression analyses, but the findings did not indicate significant temporal shifts. Therefore, our main results rely on models using pooled data across the entire period, focusing on occupational variations.

of underlying factors, including a growing demand for workers with diverse skill profiles to solve complex problems and an efficiency wage paid to job positions involving complex skills that are difficult to monitor from the employers' perspective. In contrast, the earnings returns from the same degree of within-job skill diversity in personal care and service jobs are only about one-fifth as high as those in management occupations.

Conclusion and Discussion

The modern labor market is often depicted as demanding a workforce equipped with a diverse array of intricate skills. Mapping workplace skill profiles is essential for understanding inequality in access to job opportunities, the sorting of workers into occupations and jobs, and the broader distribution of economic rewards. Despite the centrality of this topic, the literature remains rather incomplete. Previous research, primarily based on surveys and administrative data, has focused on skill profiles at the occupation level. Our study calls attention to the conceptual distinction between occupation- and job-level skill diversity. We argue that jobs, rather than occupations, are the fundamental units where specific skill sets are negotiated, performed, evaluated, and compensated. By combining large-scale data from online job postings with nationally representative surveys, our study offers the first comprehensive investigation into trends in skill diversification across the occupational spectrum.

Moreover, we challenge the prevailing assumption that the increase in skill diversity is only confined to high-paying jobs. In line with the widespread expectations, within-job skill diversity has increased in management and professional occupations. What is somewhat unexpected is that the rise in skill diversity in the personal care and service sector is three times as large as that in management occupations. These findings substantiate our hypotheses that skill diversification within jobs has extended well beyond high-paying occupations. Although possessing a wide range of skills doesn't inherently indicate skill complexity, these findings do call into question the prevailing notion that low-wage personal service jobs require only

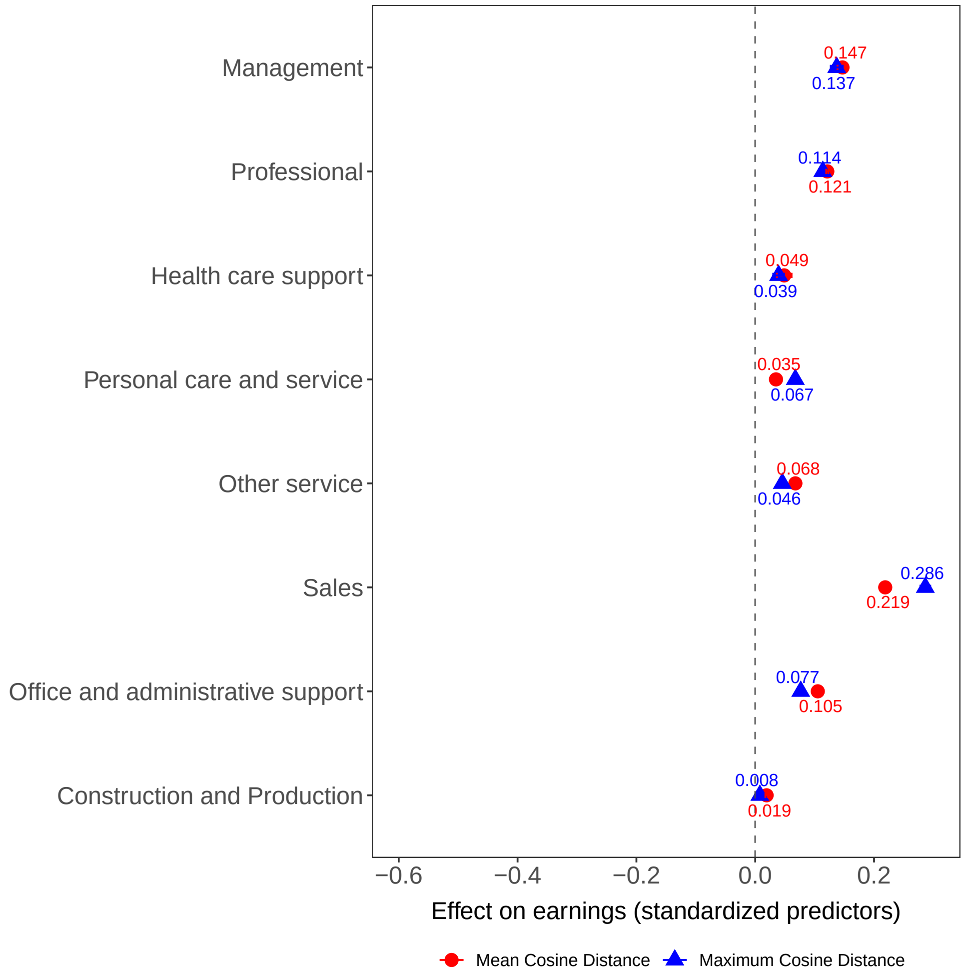


Figure 12: Earnings Returns to a One-SD Increase in Within-job Skill Diversity by Occupation Group

Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017. The measures of within-job diversity have been standardized to a mean of zero and a standard deviation of one. The outcome variable, annual earnings, is measured on a logarithmic scale.

Alt Text: Coefficient plot showing earnings returns to a one standard deviation increase in within-job skill diversity across eight occupational groups. Red circles show mean cosine distance and blue triangles show maximum cosine distance. Sales has the largest positive effect; construction and production has the smallest.

simple and homogeneous skills. In their own account, for example, care workers today are "multi-skilled emotional managers" and "jack of all trades" (Malhotra, 2022). The precarious yet flexible nature of their work allows them to absorb job tasks from other occupations, which may further diversify the skill set of their own jobs. Yet, many of their skills frequently go unnoticed and undervalued. Why, then, did the personal care and service jobs not receive much attention in the discussion of the impact of technological upgrading on skill profiles? While this is beyond the scope of this study, prior work suggests that this might be driven partly by a tendency to consider low-paying jobs as low-skill jobs; in other words, whether a job is considered skillful or not depends on the perceived social status of workers (e.g., gender, race, and ethnicity), rather than the actual skill set involved (Warhurst et al., 2017; Attewell, 1990). The current literature's disproportionate focus on skill sets in management and professional jobs could potentially overshadow the significance of diverse skills required for lower-wage jobs. This lack of attention is especially troubling for researchers studying social disparities, as our supplementary analysis revealed that the rise in within-job skill diversification in service jobs is particularly pronounced among women and racial minorities (see Figure S11 in Online Supplement).[31] In light of the sustained expansion of employment in personal care and service jobs, we call for future research to examine the evolving patterns and trends in the skill sets required for workers in these positions. Further, even when we examine trends *within* each occupational group, skill diversification is not always confined to high-paying workers. In management occupations, workers with the highest levels of education and income experienced the most pronounced increase in skill diversity. In contrast, among professional and personal care and service occupations, the most rapid diversification occurred among workers with lower education and income levels. Focusing exclusively on high-paying jobs, therefore, risks overlooking alternative forces driving skill diversification among relatively low-wage workers.

Beyond these substantive contributions, this study also offers a new methodological ap-

[31] Note that these trends do not account for the differential selectivity into different occupational groups by race and gender.

proach for addressing an important challenge in the measurement of skill diversity—namely, the incorporation of skill-to-skill relationships. We developed a relational framework that we term "Skill Embedding," which maps interrelated skills onto a multi-dimensional space to capture the interdependencies among skills. Our framework takes into account not only the direct associations between skills (e.g., skills A and B appearing in the same job) but also the indirect connections between skills through their co-occurrence patterns with other skills in the entire mapping of all skills demanded by jobs in the labor market (e.g., skills A and B often co-occurring with skill C). Our framework reveals the complex ways in which various skills are bundled and integrated into jobs and how these skills are interconnected with each other. Our empirical analysis serves as an example of how this framework can be useful for grasping the dynamics of skill profiles, and we encourage future studies to take advantage of this framework and extend this to other topics of labor market research.

Our study is not without limitations. First, the EBG database primarily encompasses job positions advertised on online platforms, potentially introducing selection bias. This bias may inadvertently exclude lower-end jobs not typically recruited online. To mitigate this, we matched the skill diversity measures from the EBG database with the nationally representative ACS data. However, this approach may still result in a more accurate depiction of skill profiles for higher-earning occupations. Similarly, the job postings may include detailed items of skill demands, but not their intensity, importance, or level of mastery. It tells only an employer-side story, showing how companies present their skill demands. However, the described skills and economic returns may differ from what eventually gets written into a worker's contract. There can also be a potential gap between the skills listed in job postings and those actually utilized in the workplace. We acknowledge these inherent limitations, but we believe our study is among the first to highlight the significant potential of online-scraped big data in addressing long-standing questions in the sociology of work and social stratification. Second, since we matched the year-specific EBG and ACS data using the SOC detailed occupational scheme, our estimates of trends and earnings returns on skill diversity are based on detailed-occupation-level average within-job skill diversity serving as a proxy for individual-level job diversity. Future research should explore additional data sources to further examine earnings effects at the individual level.

Moving forward, our skill embedding approach provides a promising foundation for exploring a broader spectrum of research questions. Our finding of a substantial surge in skill diversity within three rapidly expanding occupational groups offers a basis for tentative speculation about the impact of job polarization. Should job polarization persist in the future, one may question if it will lead to a further escalation in skill diversity within these occupations, or if this trend could expand to jobs in the middle of the earnings spectrum as well. Moreover, how does job polarization interact with other labor market dynamics, such as structural inequality between gender and racial groups (Duffy, 2011; Dwyer, 2013), to shape skill requirements and their economic returns at work? Are changes in skill requirements over time driven by shifts in the relationships between specific skill pairs? These are some of the intriguing queries to be explored in future research.

Another promising future direction involves tracking the diffusion of skill profiles of the same occupation within and across different organizations. For example, it would be interesting to investigate how the early adoption of certain skill combinations might become more widespread over time in other organizations that share similar culture, industry structures, or geographical locations. Moreover, the vector space created through our skill embedding can be utilized to examine how specific skill pairings, such as the combination of social science knowledge and programming skills, evolve over time. Given the rapid technological advancements, including the widespread integration of AI in various tech and non-tech sectors and its impact on workers and employers in these sectors, our framework, capable of capturing the dynamic relationship between skills, is set to become a critical tool for future labor market research.

Methodological Details

Part I: Mapping between Conceptual and Analytical Elements of the Skill Embedding Framework

Table A1: Mapping between Conceptual and Analytical Elements of Skill Embedding

Conceptual Element	**Analytical Element**
1. Orthogonalization of skill dimensions	SVD decomposition (V_k) of the skill-job matrix
2. Projection of skills onto a vector space	Obtaining skill vectors from SVD (matrix product of V_k and Σ_k)
3. Ordering of dimensions by importance	Diagonal cells of matrix Σ
4. Measuring skill-to-skill distance	Cosine similarity and cosine distance
5. Measuring skill diversity	Average pair-wise cosine distance

Part II: Contrasting Within-job and Between-job Skill Diversity Measures

To supplement the measure of within-job skill diversity introduced in our main text, we now contrast this within-job skill diversity measure with between-job skill diversity. To compute the latter, we first calculate the centroids of skill vectors within each job and denote them by $\tilde{v}$. Then, for an occupation o with a total number of N jobs, we define between-job skill diversity ($\delta^{between}$) as the average pair-wise cosine distance across all job-level centroid skill vectors within occupation o:

$$\text{Between-job skill diversity} = \delta^{between} = \frac{2}{N(N-1)} \Sigma_{k,l \in \Phi_o \& k \neq l} d_{\tilde{v}_k, \tilde{v}_l}, \tag{7}$$

where the constant factor in front of the summation sign is the inverse of the total number of possible job pairs ($\frac{N(N-1)}{2}$), and Φ_o denotes the set of all job-level centroid vectors in occupation o.

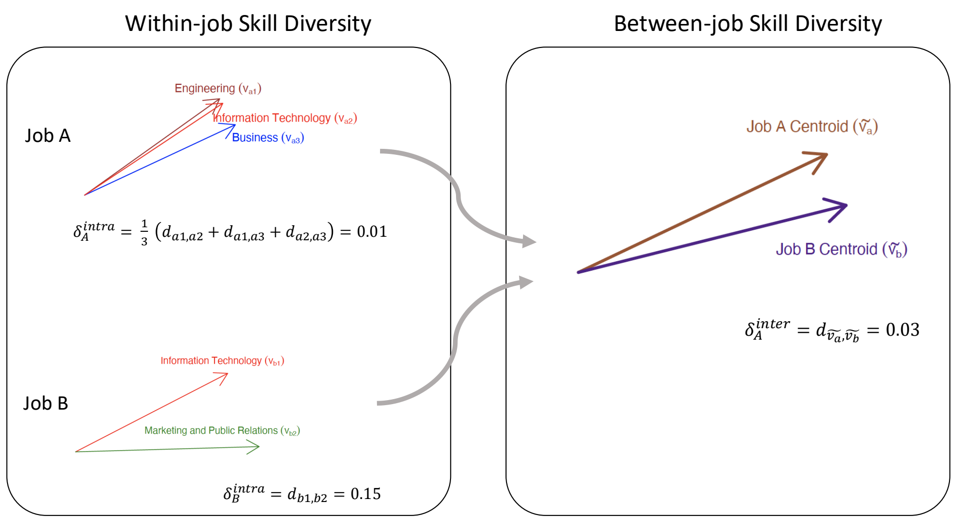


FIGURE A1: ILLUSTRATING THE DIFFERENCE BETWEEN WITHIN-JOB AND BETWEEN-JOB SKILL DIVERSITY IN THE SKILL VECTOR SPACE

Two boxed panels illustrate skill diversity computation for two hypothetical jobs. Left panel shows skill vectors within each job; Job A has three similar skill arrows (diversity = 0.01), Job B has two divergent skill arrows (diversity = 0.15). Right panel shows the two job centroid vectors; between-job diversity = 0.03.

Figure A1 provides a numerical demonstration of the distinction between δ^{within} and $\delta^{between}$, using the same pair of hypothetical jobs previously discussed in Figure 3. The left-side plot illustrates the process of determining within-job skill diversity. For Job A, we compute δ_A^{within} by finding the mean of the three potential pairwise cosine distances, $d_{a1,a2}$, $d_{a1,a3}$, and $d_{a2,a3}$, resulting in a value of 0.01. In the case of Job B, which requires only two skills, δ_B^{within} is simply the cosine distance between the two associated vectors, yielding a value of 0.15. By our measure, within-job skill diversity is higher in Job B than in Job A. On the right-hand side of the figure, the focus shifts to the computation of between-job skill diversity. This is achieved by first generating the centroid vectors for the two jobs, denoted as $\tilde{v}_a$ and $\tilde{v}_b$, followed by calculating $\delta^{between}$ as the cosine distance between these centroid vectors, arriving at a value of 0.03.

Part III: Examples of Original Job Postings with Varying Degrees of Skill Diversity

To gain a better understanding of job postings data, we show how skills are described in the original online job postings. Our goal is to highlight the varying degrees of skill diversity that can be evident in such postings. To achieve this, we will provide three exemplary job descriptions derived directly from the raw text of job postings data and showcase the skills that have been extracted from the EBG dataset. First, we present a job posting titled "Before and After School Teacher needed for School-Age Children." This position requires a comparatively homogeneous set of skills centered around child care and education:

> ***EBG Example Job #1: Before and After School Teacher needed for School-Age Children (Waltham, Massachusetts)***
> ***Description:*** *Join our talented team, where we inspire children to be lifelong learners Through our play based curriculum, our affectionate and loving staff ensures that our children are imparted with the knowledge to succeed. Our Lead Teachers... Create fun and interactive learning experiences while serving as mentors to fellow Teachers. Are caring, compassionate and love what they do Ensure the daily care of every child by following all licensing guidelines and implement-*

ing all company standards. Communicate directly with parents and prospective parents to achieve success for the child. Maintain a fun and interactive classroom that is clean and organized. Have countless advancement opportunities through our on-going training and expansive network of centers and brands. Are rewarded with hugs from children and praise from parents every day We are looking for candidates that are as passionate about the growth and development of the precious children in our care as we are. We are most interested in talking to applicants that have: Experience leading a classroom and creating educational lesson plans Experience working in a licensed childcare facility Coursework or a degree in early childhood education or child development or a CDA Impeccable references and a proven track record of caring and nurturing children to provide them with a great start to their educational careers The ability to meet state and/or accreditation requirements for education and experience Flexibility as to the hours and schedule of work Must be at least 18 years of age.

EBG Extracted Skills*: Teaching, Child Development, Child Care, Childhood Education and Development*

Next, we provide a job posting for a "Biological Field Technician/Supervisor," a role demanding a more diverse array of skills. As per the skill categories extracted, this position requires expertise in scientific research, encompassing areas such as biology, data techniques, and ecology. Additionally, this job requires skills in resource management and a foundational level of customer service skills, showcasing a broader and more varied skill set:

EBG Example Job #2: Biological Field Technician/Supervisor (Northeast or Mid-Atlantic U.S.)
Job Description: *Primary responsibilities include conducting field work, supervising, and training field technicians. Successful candidates must be able to live and work in a remote field setting; be willing to travel; work well both independently and with colleagues; follow field, safety, and data collection protocols; and follow instruction of supervisors. Job duties will include field set-up and surveys for bats and birds at proposed wind-energy projects throughout the northeast and mid-Atlantic. Previous experience conducting acoustic or mist-net surveys for bats, and point count surveys for birds and raptors is important. Job duties require use of acoustic bat detectors, global positioning system devices, computers and digital cameras. An average work week may include 30-50 hours, though some weeks may require shorter or longer hours, depending on seasonal workloads and weather conditions. Housing and field vehicle will be provided for travel outside of the designated home station. Field equipment will be provided by WEST; however, field technicians must be equipped with suitable footwear (rugged hiking boots or steel-toed boots) and appropriate field clothing.*

> ***EBG Extracted Skills****: Science and Research: Biology, Data Techniques, Ecology, Resource Management and Restoration, Basic Customer Service.*

Finally, we present a job posting for a Gift Shop Manager, a position that requires a considerably diverse skill set. This job calls for a wide range of management-related abilities alongside several administrative competencies, including scheduling, foundational customer service, and employee relations. Furthermore, it demands industry-specific expertise in areas like retail sales, retail store operations, and procurement, showcasing the multifaceted and diverse nature of the skill profile required for this particular job:

> ***EBG Example Job # 3: Gift Shop Manager (Scotrun, Pennsylvania)***
> ***Job Description:*** *Willingness to accept the most effective role. Responsible for the strategic planning and development of the department. Responsible for the daily operation of all retail areas including gift shops, arcade, vending, and any other revenue producing venues. Oversees all merchandising functions including store display and design, inventory, tracking/analyzing, merchandise purchasing, par levels and pricing, merchandise/materials sourcing. Develops and manages labor, COS, and expense budgets. Manages staff including employee-training programs, scheduling and employee relations. Oversees sales functions including POS system, cash handling, budgets and daily reporting. Ensures retail areas achieve the highest standards in areas of cleanliness, presentation and service. Develops, implements, and monitors programs assuring a safe facility and work environment that is incompliance with all appropriate regulations Ergonomics, Emergency Responses, Injury and Illness Prevention, and Hazard Communications Programs. Orders and maintains supply levels. Ensures retail areas achieve the highest standards in areas of cleanliness, presentation and service.*
>
> ***EBG Extracted Skills****: Budget Management, People Management, Physical Abilities, Merchandising, Basic Customer Service, Retail Store Operations, Cash Register Operation, Business Communications, Supplier Relationship Management, Scheduling, Business Strategy, Employee Relations, Retail Sales, Procurement, Training Programs.*

Part IV: Details on Skill Categories in EBG Data

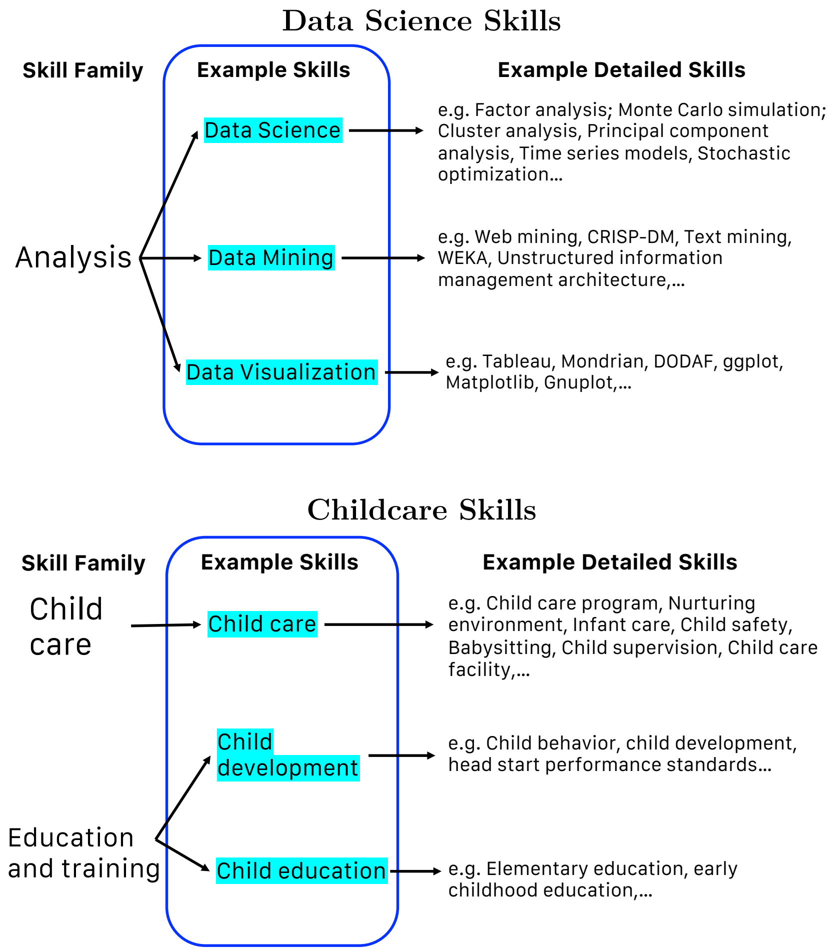


Figure A2: Examples of Skill Categories in EBG

Alt Text: Two hierarchical diagrams illustrating the EBG skill taxonomy. Top: "Analysis" cluster family branches into example skills Data Science, Data Mining, and Data Visualization, each with detailed examples. Bottom: "Education and Training" family branches into Child Care, Child Development, and Child Education, each with detailed example skills listed.

References

Alabdulkareem, Ahmad, Morgan R Frank, Lijun Sun, Bedoor AlShebli, César Hidalgo, and Iyad Rahwan (2018). "Unpacking the Polarization of Workplace Skills". In: *Science Advances* 4.7, eaao6030.

Althobaiti, Shahad, Ahmad Alabdulkareem, Judy Hanwen Shen, Iyad Rahwan, Morgan Frank, Esteban Moro, and Alex Rutherford (2022). "Longitudinal Complex Dynamics of Labour Markets Reveal Increasing Polarisation". In: *arXiv e-prints*. DOI: 10.48550/arXiv.2204.07073. eprint: 2204.07073.

Altonji, Joseph G. (2010). *Multiple Skills, Multiple Types of Education, and the Labor Market: A Research Agenda*. DOI: 10.2139/ssrn.1888514.

Anderson, Katharine A (2017). "Skill Networks and Measures of Complex Human Capital". In: *Proceedings of the National Academy of Sciences* 114.48, pp. 12720–12724.

Attewell, Paul (1990). "What is Skill?" In: *Work and Occupations* 17.4, pp. 422–448. DOI: 10.1177/0730888490017004003.

Autor, David H and David Dorn (2013). "The growth of low-skill service jobs and the polarization of the US labor market". In: *American Economic Review* 103.5, pp. 1553–1597.

Autor, David H, Lawrence F Katz, and Melissa S Kearney (2008). "Trends In US Wage Inequality: Revising The Revisionists". In: *The Review Of Economics And Statistics* 90.2, pp. 300–323.

Avent-Holt, Dustin, Lasse Folke Henriksen, Anna Erika Hägglund, Jiwook Jung, Naomi Kodama, Silvia Maja Melzer, Eunmi Mun, Anthony Rainey, and Donald Tomaskovic-Devey (2020). "Occupations, workplaces or jobs?: An exploration of stratification contexts using administrative data". In: *Research in Social Stratification and Mobility* 70, p. 100456.

Banerjee, Supurna (2023). "Laboring Femininities: Skill, Body, and Class-making Among Beauty Workers in India". In: *International Labor and Working-Class History* 104, pp. 77–102.

Barney, Jay B. and Patrick M. Wright (1998). "On Becoming a Strategic Partner: The Role of Human Resources in Gaining Competitive Advantage". In: *Human Resource Management* 37.1, pp. 31–46. ISSN: 0090-4848.

Baron, James N and William T Bielby (1980). "Bringing the firms back in: Stratification, segmentation, and the organization of work". In: *American Sociological Review*, pp. 737–765.

— (1986). "The Proliferation of Job Titles in Organizations". In: *Administrative Science Quarterly*, pp. 561–586.

Baron, James N and Andrew E Newman (1990). "For What It's Worth: Organizations, Occupations, and the Value of Work Done by Women and Nonwhites". In: *American Sociological Review*, pp. 155–175.

Bauer, Lauren and Sarah Yu Wang (2023). *Prime-Age Women Are Going Above and Beyond in the Labor Market Recovery*.

Becker, Gary S (1962). "Investment in Human Capital: A Theoretical Analysis". In: *Journal of Political Economy* 70.5, Part 2, pp. 9–49.

Blau, Francine D, Peter Brummund, and Albert Yung-Hsu Liu (2013). "Trends in Occupational Segregation by Gender 1970–2009: Adjusting for the Impact of Changes in the Occupational Coding System". In: *Demography* 50.2, pp. 471–492.

Blau, Peter M., Otis Dudley Duncan, and Andrea Tyree (1978). *American Occupational Structure*. Free Press.

Bol, Thijs, Christina Ciocca Eller, Herman G Van De Werfhorst, and Thomas A DiPrete (2019). "School-to-Work Linkages, Educational Mismatches, and Labor Market Outcomes". In: *American Sociological Review* 84.2, pp. 275–307.

Bolton, Sharon C. (2004). *Conceptual Confusions: Emotion Work as Skilled Work*. London, England, UK: Palgrave Macmillan. ISBN: 978-140390639.

Börner, Katy, Olga Scrivner, Mike Gallant, Shutian Ma, Xiaozhong Liu, Keith Chewning, Lingfei Wu, and James A Evans (2018). "Skill Discrepancies between Research, Education, and Jobs Reveal the Critical Need to Supply Soft Skills for the Data Economy". In: *Proceedings of the National Academy of Sciences* 115.50, pp. 12630–12637.

Brynjolfsson, Erik (2014). *The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies*. W. W. Norton & Company.

Bryson, Jane (2017). "Disciplinary Perspectives on Skill". In: *The Oxford Handbook of Skills and Training*. Ed. by John Buchanan et al. Oxford: Oxford University Press, pp. 17–35. ISBN: 9780199655366. DOI: 10.1093/oxfordhb/9780199655366.013.1.

Burt, Ronald S (1995). *Structural Holes: The Social Structure of Competition*. Harvard University Press.

Byun, Heejung and Joseph Raffiee (2023). "Career Specialization, Involuntary Worker–Firm Separations, and Employment Outcomes: Why Generalists Outperform Specialists When Their Jobs Are Displaced". In: *Administrative Science Quarterly* 68.1, pp. 270–316.

Carroll, Joseph (1995). "Evolution and Literary Theory". In: *Human Nature* 6, pp. 119–134.

Cattell, Raymond B (1966). "The Scree Test for the Number of Factors". In: *Multivariate Behavioral Research* 1.2, pp. 245–276.

Cheng, Siwei, Bhumika Chauhan, and Swati Chintala (2019). "The Rise Of Programming And The Stalled Gender Revolution". In: *Sociological Science* 6, pp. 321–351.

Cheng, Siwei, Andrew Levine, and Ananda Martin-Caughey (2025). "Using Relative Distribution Methods to Study Economic Polarization across Categories and Contexts". In: *Sociological Methodology* 55.1, pp. 91–120. DOI: 10.1177/00811750241260731.

Clegg, Chris and Caroline Spencer (2007). "A Circular and Dynamic Model of the Process of Job Design". In: *Journal of Occupational and Organizational Psychology* 80.2, pp. 321–339. ISSN: 0963-1798. DOI: 10.1348/096317906X113211.

Deming, David and Lisa B Kahn (2018). "Skill Requirements across Firms and Labor Markets: Evidence from Job Postings for Professionals". In: *Journal of Labor Economics* 36.S1, S337–S369.

Deming, David J (2017). "The Growing Importance Of Social Skills In The Labor Market". In: *The Quarterly Journal Of Economics* 132.4, pp. 1593–1640.

DiPrete, Thomas A and Joanna Chae (2023). "Coordinated Markets, School-to-work Linkages, and Labor Market Outcomes in Europe". In: *Research in Social Stratification and Mobility* 87, p. 100840.

Duffy, Mignon (2011). *Making Care Count: A Century of Gender, Race, and Paid Care Work*. Rutgers University Press.

Dwyer, Rachel E (2013). "The care economy? Gender, economic restructuring, and job polarization in the US labor market". In: *American Sociological Review* 78.3, pp. 390–416.

Eggenberger, Christian, Miriam Rinawi, and Uschi Backes-Gellner (2018). "Occupational Specificity: A New Measurement Based on Training Curricula and Its Effect on Labor Market Outcomes". In: *Labour Economics* 51, pp. 97–107.

England, Paula (2005). "Emerging Theories of Care Work". In: *Annual Review of Sociology* 31, pp. 381–399.

England, Paula, Andrew Levine, and Emma Mishel (2020). "Progress toward gender equality in the United States has slowed or stalled". In: *Proceedings of the National Academy of Sciences* 117.13, pp. 6990–6997.

Faulkner, Robert R (1983). *Music on Demand*. Transaction Publishers.

Gatta, Mary, Heather Boushey, and Eileen Appelbaum (2009). "High-Touch and Here-to-Stay: Future Skills Demands in US Low Wage Service Occupations". In: *Sociology* 43.5, pp. 968–989. ISSN: 0038-0385. DOI: 10.1177/0038038509340735.

Gibbons, Robert and Michael Waldman (2004). "Task-specific human capital". In: *American Economic Review* 94.2, pp. 203–207.

Goldin, Claudia and Lawrence F Katz (2010). *The Race Between Education And Technology.* Harvard University Press.

Gonalons-Pons, Pilar and Ioana Marinescu (2022). *Care Labor Demand Shocks and Inequality: How Childcare Costs Exacerbate Inequality among American Families.* Working Paper.

Han, Siqi, Jack LaViolette, Chad Borkenhagen, William McAllister, and Peter S Bearman (2023). "Interdisciplinary college curriculum and its labor market implications". In: *Proceedings of the National Academy of Sciences* 120.43, e2221915120.

Hannan, Michael T and John Freeman (1977). "The Population Ecology of Organizations". In: *American Journal of Sociology* 82.5, pp. 929–964.

Hauser, Robert M and John Robert Warren (1997). "Socioeconomic Indexes for Occupations: A Review, Update, and Critique". In: *Sociological Methodology* 27.1, pp. 177–298.

Hénaut, Léonie, Jennifer C Lena, and Fabien Accominotti (2023). "Polyoccupationalism: Expertise Stretch and Status Stretch in the Postindustrial Era". In: *American Sociological Review* 88.5, pp. 872–900.

Hershbein, Brad and Lisa B Kahn (2018). "Do Recessions Accelerate Routine-biased Technological Change? Evidence from Vacancy Postings". In: *American Economic Review* 108.7, pp. 1737–1772.

Hershbein, Brad J. and Kevin M. Hollenbeck (2014). *Refining Workforce Education Supply and Demand Analysis: Final Report.* Upjohn Institute Technical Report 15-031. Kalamazoo, MI: W.E. Upjohn Institute for Employment Research. DOI: 10.17848/tr15-031. URL: https://research.upjohn.org/up_technicalreports/31.

Hosseinioun, Moh, Frank Neffke, Letian Zhang, and Hyejin Youn (2025). "Skill Dependencies Uncover Nested Human Capital". In: *Nature Human Behavior* 9, pp. 673–687. ISSN: 2397-3374. DOI: 10.1038/s41562-024-02093-2.

Hout, Michael (1984). "Status, Autonomy, and Training in Occupational Mobility". In: *American Journal of Sociology* 89.6, pp. 1379–1409.

Huang, Ming-Hui and Roland T Rust (2018). "Artificial Intelligence in Service". In: *Journal of Service Research* 21.2, pp. 155–172.

Jackson, Michelle (2025). *The Division of Rationalized Labor*. Harvard University Press.

Juraschek, Stephen P., Xiaoming Zhang, Vinoth Ranganathan, and Vernon W. Lin (2019). "United States Registered Nurse Workforce Report Card and Shortage Forecast". In: *American Journal of Medical Quality* 34.5, pp. 473–481. ISSN: 1555-824X. DOI: 10.1177/1062860619873217. eprint: 31479295.

Kalleberg, Arne L (2009). "Precarious Work, Insecure Workers: Employment Relations in Transition". In: *American Sociological Review* 74.1, pp. 1–22.

Kalleberg, Arne L. (2011). *Good Jobs, Bad Jobs*. Russell Sage Foundation.

Kalleberg, Arne L and Ted Mouw (2018). "Occupations, Organizations, and Intragenerational Career Mobility". In: *Annual Review of Sociology* 44, pp. 283–303.

Khaouja, Imane, Ismail Kassou, and Mounir Ghogho (2021). "A Survey on Skill Identification from Online Job Ads". In: *IEEE Access* 9, pp. 118134–118153.

Kim, ChangHwan and Arthur Sakamoto (2008). "The Rise of Intra-Occupational Wage Inequality in the United States, 1983 to 2002". In: *American Sociological Review* 73.1, pp. 129–157.

Kornrich, Sabino and Frank Furstenberg (2013). "Investing in Children: Changes in Parental Spending on Children, 1972–2007". In: *Demography* 50.1, pp. 1–23.

Lancaster, Vicki, Devika Mahoney-Nair, and Nathaniel J Ratcliff (2019). "Technology Report Review of Burning Glass Job-ad Data". In: *Biocomplexity Institute and Initiative Social and Decision Analytics Division, University of Virginia*.

Lehdonvirta, Vili (2018). "Flexibility in the Gig Economy: Managing Time on Three Online Piecework Platforms". In: *New Technology, Work and Employment* 33.1, pp. 13–29.

Leung, Ming D (2014). "Dilettante or Renaissance Person? How the Order of Job Experiences Affects Hiring in an External Labor Market". In: *American Sociological Review* 79.1, pp. 136–158.

Liu, Yujia and David B Grusky (2013). "The Payoff To Skill In The Third Industrial Revolution". In: *American Journal of Sociology* 118.5, pp. 1330–1374.

Malhotra, Ragini Saira (2022). "Not Everyone Can Do This: Childcare Context and the Practice of Skill in Emotional Labor". In: *International Journal of Child Care and Education Policy* 16.1, p. 8.

Martin-Caughey, Ananda (2021). "What's in an Occupation? Investigating Within-Occupation Variation and Gender Segregation Using Job Titles and Task Descriptions". In: *American Sociological Review* 86.5, pp. 960–999.

Merluzzi, Jennifer and Damon J Phillips (2016). "The Specialist Discount: Negative Returns for MBAs with Focused Profiles in Investment Banking". In: *Administrative Science Quarterly* 61.1, pp. 87–124.

Modestino, Alicia Sasser, Mary A. Burke, Shahriar Sadighi, Rachel Sederberg, Tomer Stern, and Bledi Taska (2026). "No Longer Qualified? Changes in the Supply and Demand for Skills within Occupations". In: *ILR Review* 79.3, pp. 486–522. DOI: 10.1177/00197939251413182.

Morgeson, Frederick P. and Stephen E. Humphrey (2006). "The Work Design Questionnaire (WDQ): Developing and Validating A Comprehensive Measure for Assessing Job Design and the Nature of Work". In: *Journal of Applied Psychology* 91.6, pp. 1321–39. ISSN: 0021-9010. DOI: 10.1037/0021-9010.91.6.1321.

Oldham, Greg R. and J. Richard Hackman (2010). "Not What It Was and Not What It Will Be: The Future of Job Design Research". In: *Journal of Organizational Behavior* 31.2-3, pp. 463–479. ISSN: 0894-3796. DOI: 10.1002/job.678.

Parkin, Frank (1982). “Social Closure and Class Formation”. In: *Classes, Power, and Conflict*. Springer, pp. 175–184.

Piketty, Thomas and Emmanuel Saez (2006). “The Evolution of Top Incomes: A Historical and International Perspective”. In: *American Economic Review* 96.2, pp. 200–205.

Poletaev, Maxim and Chris Robinson (2008). “Human Capital Apecificity: Evidence from the Dictionary of Occupational Titles and Displaced Worker Surveys, 1984–2000”. In: *Journal of Labor Economics* 26.3, pp. 387–420.

Reckrey, Jennifer M, Emma K Tsui, R Sean Morrison, Emma T Geduldig, Robyn I Stone, Katherine A Ornstein, and Alex D Federman (2019). “Beyond Functional Support: The Range of Health-related Tasks Performed in the Home by Paid Caregivers in New York”. In: *Health Affairs* 38.6, pp. 927–933.

Saari, Margaret, Sarah Xiao, Alissa Rowe, Erin Patterson, Tieghan Killackey, Julia Raffaghello, and Ann E. Tourangeau (2018). “The role of unregulated care providers in home care: A scoping review”. In: *Journal of Nursing Management* 26.7, pp. 782–794. ISSN: 0966-0429. DOI: 10.1111/jonm.12613.

Sassen, Saskia (2001). “Impacts of Information Technologies on Urban Economies and Politics”. In: *International Journal of Urban and Regional Research* 25.2, pp. 411–418.

Schneider, Daniel and Kristen Harknett (2021). “Hard Times: Routine Schedule Unpredictability and Material Hardship among Service Sector Workers”. In: *Social Forces* 99.4, pp. 1682–1709.

Schneider, Daniel, Orestes P Hastings, and Joe LaBriola (2018). “Income inequality and class divides in parental investments”. In: *American Sociological Review* 83.3, pp. 475–507.

Snower, Dennis J. and Dennis Goerlich (2013). “Multitasking and Wages”. In: *Bonn: Institute for the Study of Labor (IZA)*. URL: https://www.econstor.eu/handle/10419/80546.

Tomaskovic-Devey, Donald and Dustin Avent-Holt (2019). *Relational Inequalities: An Organizational Approach*. Oxford University Press.

Tong, Di, Lingfei Wu, and James A. Evans (2025). "Lower-Skilled Occupations Face Greater Upskilling Pressure in U.S. Job Ads". In: *Nature Communications* 17.1237, p. 1237. DOI: 10.1038/s41467-025-67992-y.

Uzzi, Brian, Satyam Mukherjee, Michael Stringer, and Ben Jones (2013). "Atypical Combinations and Scientific Impact". In: *Science* 342.6157, pp. 468–472.

Vallas, Steven and Juliet B Schor (2020). "What Do Platforms Do? Understanding the Gig Economy". In: *Annual Review of Sociology* 46, pp. 273–294.

Vallas, Steven Peter (1990). "The Concept of Skill: A Critical Review". In: *Work and Occupations* 17.4, pp. 379–398.

Warhurst, Chris, Chris Tilly, and Mary Gatta (2017). "A New Social Construction of Skill". In: *The Oxford Handbook of Skills and Training*. Oxford University Press. ISBN: 9780199655366. eprint: https://academic.oup.com/book/0/chapter/211706311/chapter-ag-pdf/44598756/book_27995_section_211706311.ag.pdf.

Weeden, Kim A (2002). "Why Do Some Occupations Pay More Than Others? Social Closure and Earnings Inequality in the United States". In: *American Journal of Sociology* 108.1, pp. 55–101.

Weeden, Kim A and David B Grusky (2005). "The Case for a New Class Map". In: *American Journal of Sociology* 111.1, pp. 141–212.

Wilmers, Nathan (2020). "Job Turf or Variety: Task Structure as a Source of Organizational Inequality". In: *Administrative Science Quarterly* 65.4, pp. 1018–1057.

Wilmers, Nathan and Letian Zhang (2022). "Values and Inequality: Prosocial Jobs and the College Wage Premium". In: *American Sociological Review* 87.3, pp. 415–442.

Wilson, Eli R (2018). "Stuck behind kitchen doors? Assessing the work prospects of latter-generation Latino workers in a Los Angeles restaurant". In: *Ethnic and Racial Studies* 41.2, pp. 210–228.

Zuckerman, Ezra W, Tai-Young Kim, Kalinda Ukanwa, and James Von Rittmann (2003). "Robust Identities or Nonentities? Typecasting in the Feature-Film Labor Market". In: *American Journal of Sociology* 108.5, pp. 1018–1074.

Online Supplement

To
Skill Diversification Beyond High-paying Jobs

Part A Detailed Occupations in "Personal Care and Service" Group.

First-Line Supervisors of Gaming Workers; Gaming Supervisors; Slot Supervisors; First-Line Supervisors of Personal Service Workers; Animal Trainers; Nonfarm Animal Caretakers; Gaming Services Workers; Gaming Dealers; Gaming and Sports Book Writers and Runners; Gaming Service Workers, All Other; Motion Picture Projectionists; Ushers, Lobby Attendants, and Ticket Takers; Miscellaneous Entertainment Attendants and Related Workers; Amusement and Recreation Attendants; Costume Attendants; Locker Room, Coatroom, and Dressing Room Attendants; Entertainment Attendants and Related Workers, All Other; Embalmers; Funeral Attendants; Morticians, Undertakers, and Funeral Directors; Barbers, Hairdressers, Hairstylists and Cosmetologists; Barbers; Hairdressers, Hairstylists, and Cosmetologists; Miscellaneous Personal Appearance Workers; Makeup Artists, Theatrical and Performance; Manicurists and Pedicurists; Shampooers; Skincare Specialists; Baggage Porters, Bellhops, and Concierges; Baggage Porters and Bellhops; Concierges; Tour and Travel Guides; Tour Guides and Escorts; Travel Guides; Childcare Workers; Personal Care Aides; Recreation and Fitness Workers; Fitness Trainers and Aerobics Instructors; Recreation Workers; Residential Advisors; Miscellaneous Personal Care and Service Workers; Personal Care and Service Workers, All Other.

Part B Appendix Figures and tables

Figure S1: Analysis Pipeline of the Skill Embedding Framework

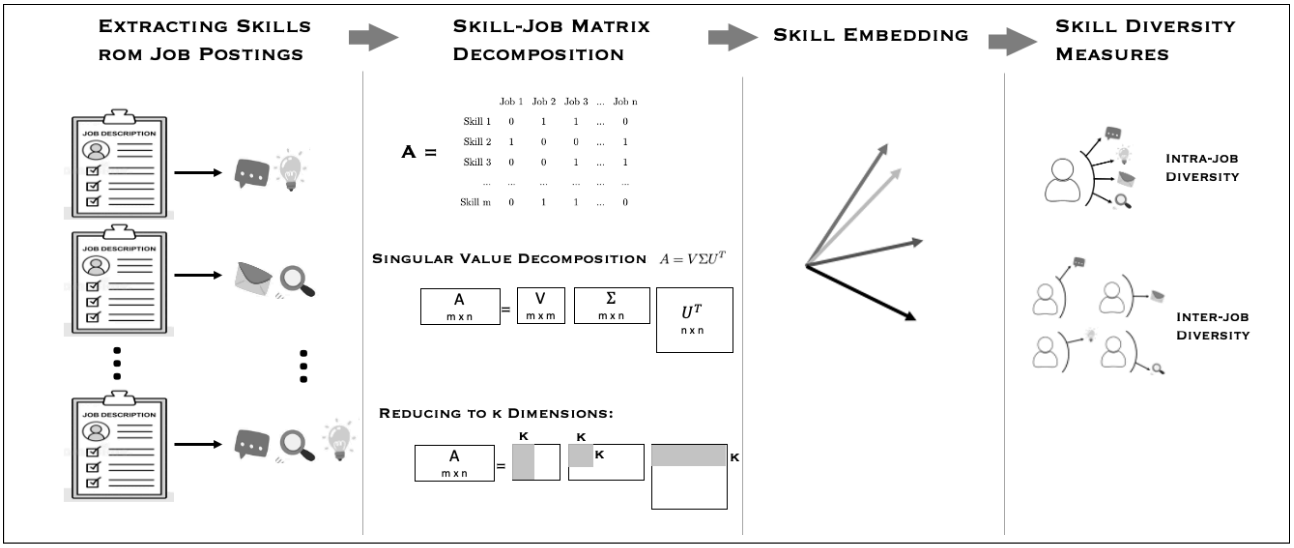
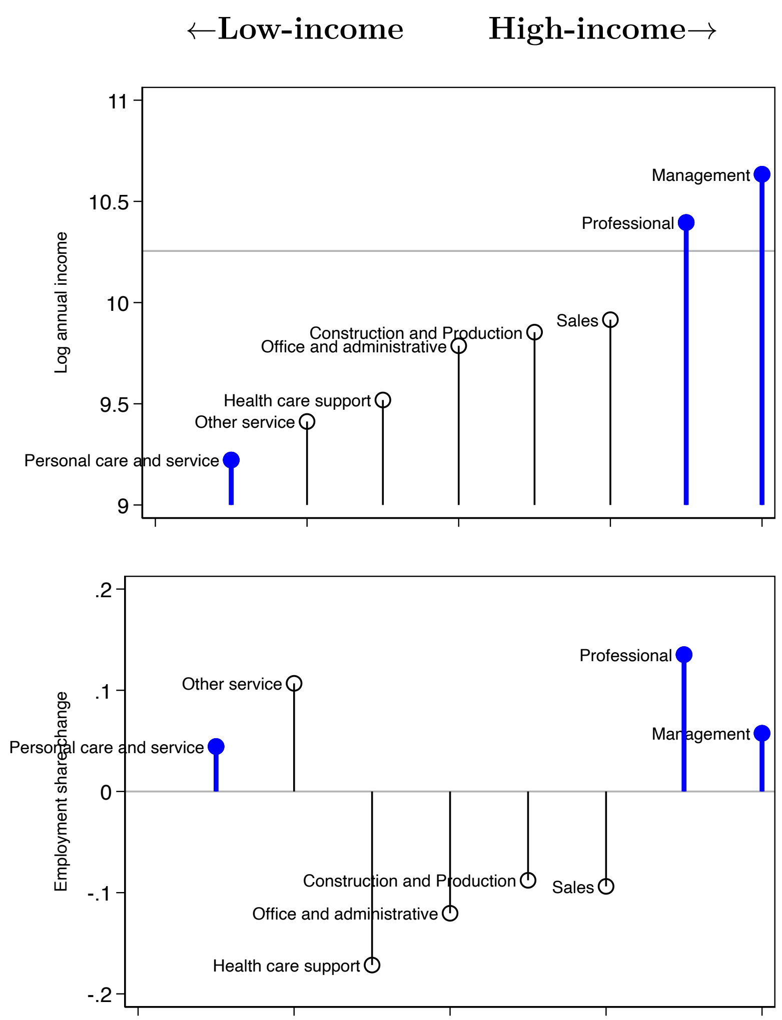
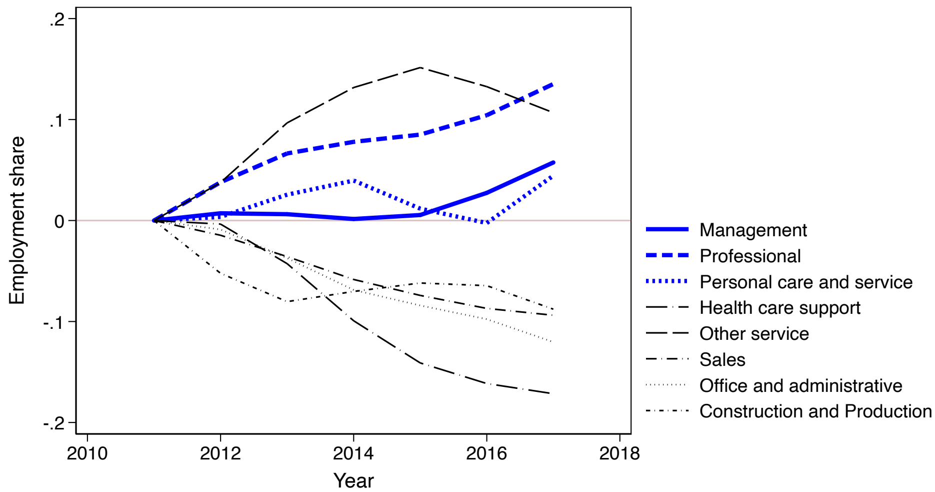


Figure S2: Mean Annual Income and Employment Share Change across Occupational Groups



Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017.

Figure S3: Percentage Increase of Employment Share from 2010



Note: Estimates are based on the American Community Survey from 2010 to 2017.

Figure S4: Histogram of Number of Skills Mentioned in Each Job Posting

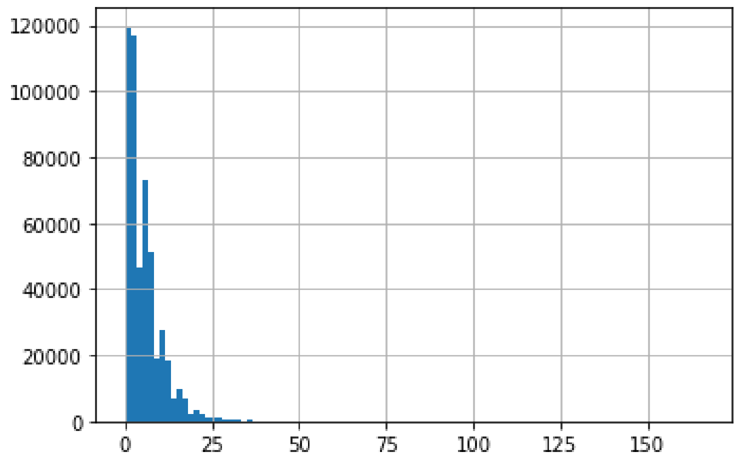


Figure S5: Number of Skills Not Included in the Common Set of Skills by Years

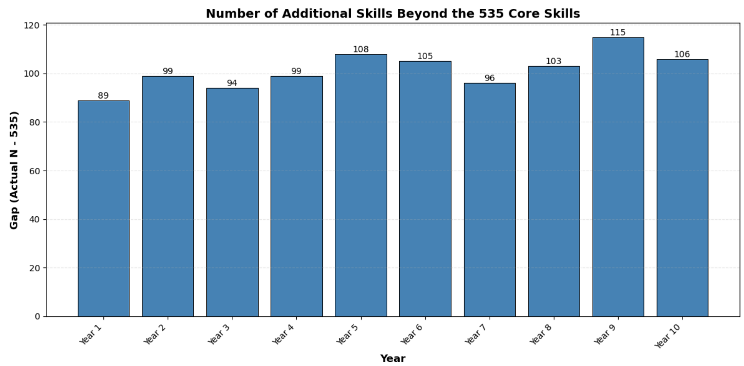


Figure S6: Scree Plot Showing the Sorted Eigenvalues in the Skill Space

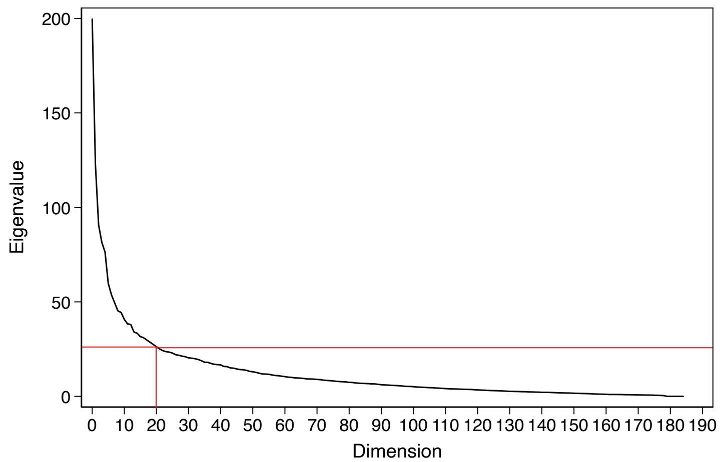


Figure S7: Comparing Employment Shares of Different Occupational Groups in EBG and ACS Data

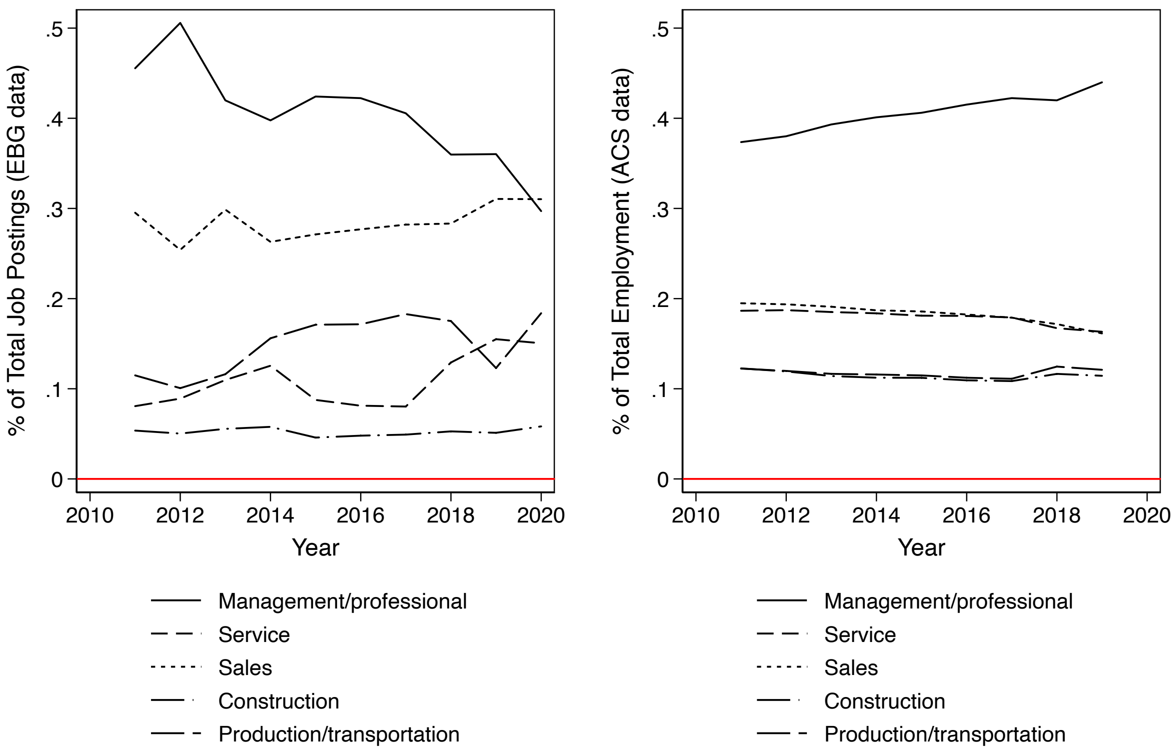


Figure S8: Trends in Within-job Skill Diversity by Disaggregated Occupations within the Professional Group

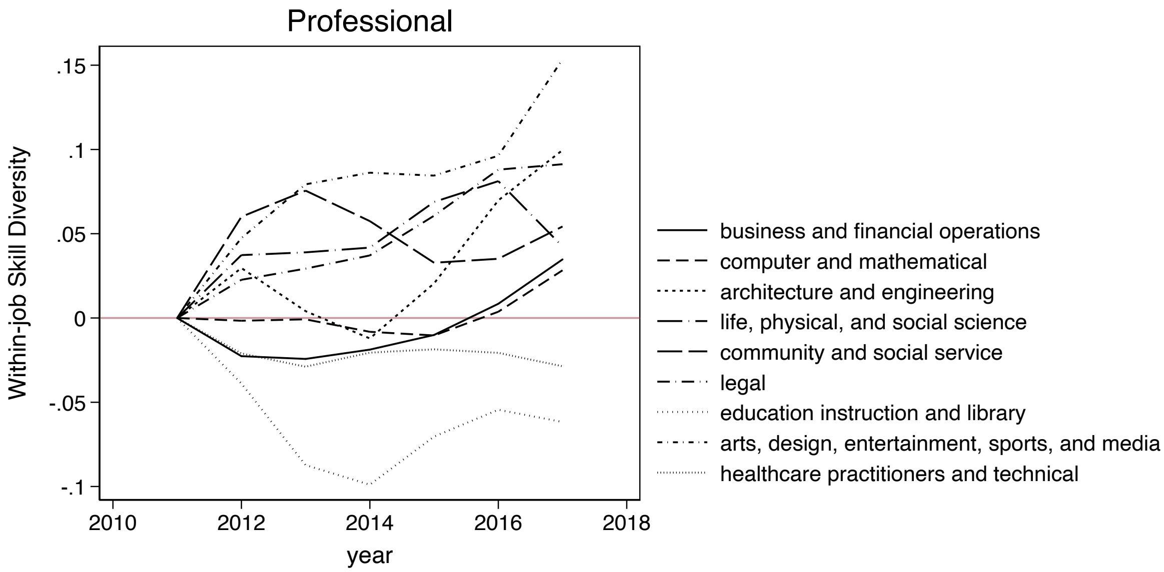


Figure S9: Trends in Within-job Skill Diversity Using Skill Embedding Trained on Re-weighted EBG Sample

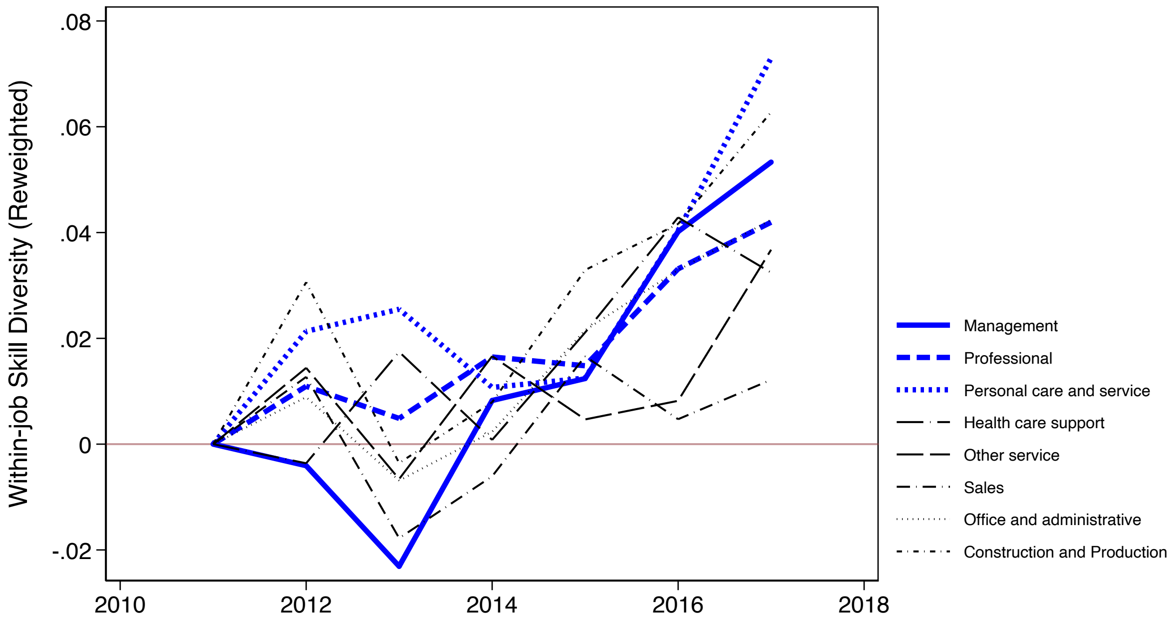


Figure S10: Trends in Within-job Skill Diversity Using Skill Embedding Trained on the 10-year Combined Sample

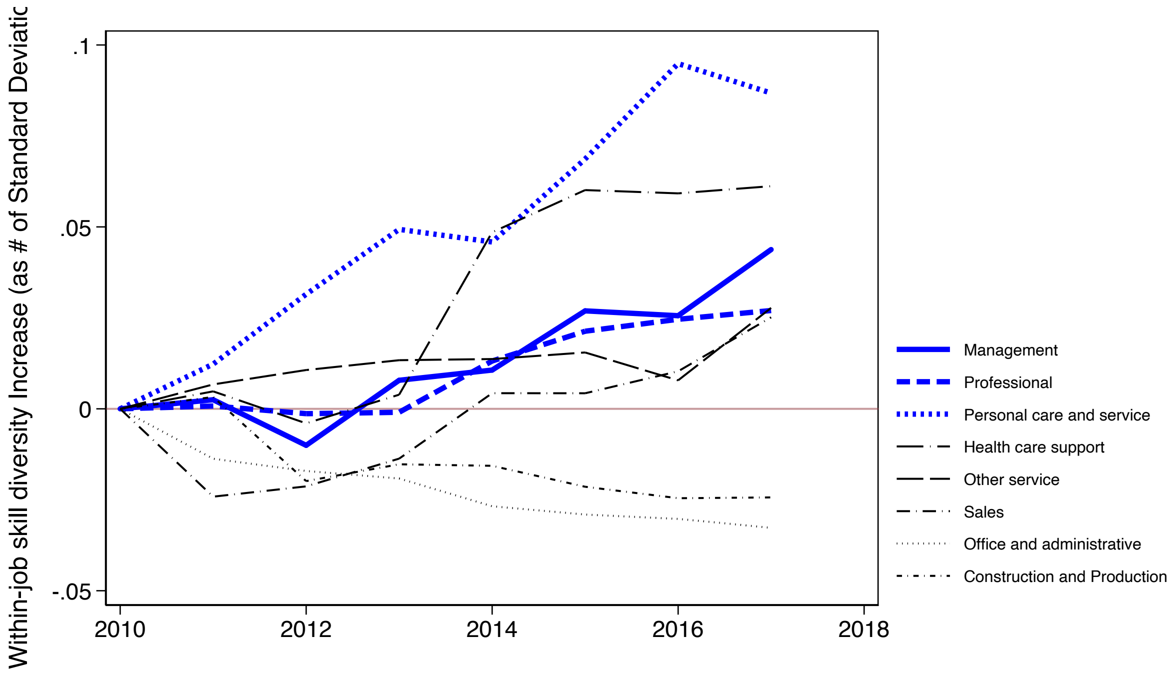
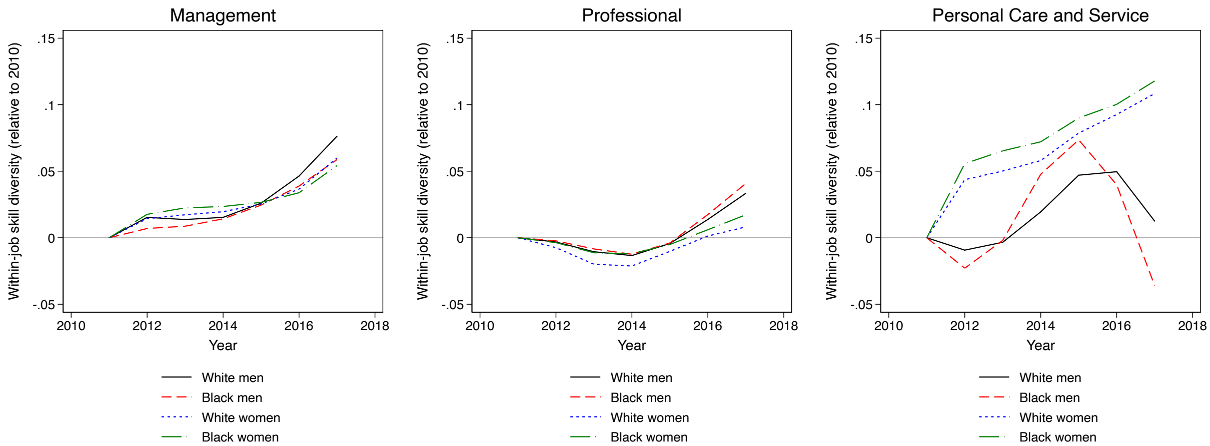


Figure S11: Trends in Within-job Skill Diversity by Gender-Race Subgroups



Note: Estimates are based on Emsi Burning Glass job postings data and the American Community Survey from 2010 to 2017.

Table S1: Monthly Sample Sizes from 2011 to 2020, with Totals and 0.1% sample.

Month	**2011**	**2012**	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**
1	1190929	1365181	2031129	2072963	2029853	2353499	2265290	2218823	2813571	3586630
2	1082027	1390795	1521442	1933468	1702752	2054328	2300116	2100508	2685178	3529201
3	1481199	1525187	1617963	2314584	2110129	2232143	2656364	2547110	3046695	3435944
4	1141400	1339037	1461004	2183930	2132652	2143602	2432853	2567904	2659102	2336034
5	1483885	1446033	1519038	2194454	2549454	2090041	2431854	3504558	3103341	2523226
6	1298601	1285551	1981526	1969086	2452501	2335961	2443332	2453778	2595144	3053210
7	1402314	1157706	2114686	1741892	2703035	2685153	2446909	3438286	2338549	3065428
8	1326937	1072471	2021130	2116676	2524029	2239998	2420213	3033214	3361369	3136598
9	1286753	1070075	1982926	2032930	2312450	2453865	2145968	2983474	3374970	2954359
10	1289154	1127952	2066091	2243279	2319190	2372467	2154025	3719062	3260927	3319012
11	1012232	1112734	1759677	1788928	1953363	2096865	1840863	2936797	3006170	2759358
12	1113216	1196271	1565545	1668683	1819082	1988581	1632061	2701206	3242094	2771652
Total	15108647	15088993	21642157	24260873	26608490	27046503	27169848	34204720	35487110	36470652
0.1% sample	15109	15089	21642	24261	26608	27047	27170	34205	35487	36470
								0.1% Sample Size = 263088		